



US012403248B2

(12) **United States Patent**
Coleman et al.

(10) **Patent No.:** **US 12,403,248 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **NEEDLE HUB FOR DRUG DELIVERY
DEVICE**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 446 days.

(21) Appl. No.: **17/569,187**

(22) Filed: **Jan. 5, 2022**

(65) **Prior Publication Data**

US 2022/0211939 A1 Jul. 7, 2022

Related U.S. Application Data

(60) Provisional application No. 63/134,054, filed on Jan.
5, 2021.

(51) **Int. Cl.**

A61M 5/142 (2006.01)

A61M 5/158 (2006.01)

A61M 5/162 (2006.01)

(52) **U.S. Cl.**

CPC **A61M 5/14248** (2013.01); **A61M 5/14244**
(2013.01); **A61M 5/158** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC **A61M 5/3293**; **A61M 5/14248**; **A61M**
5/1454; **A61M 5/158**; **A61M 5/162**;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,284,077 A 8/1981 Wagner
5,267,963 A 12/1993 Bachynsky
(Continued)

FOREIGN PATENT DOCUMENTS

JP 2002528234 A 9/2002
JP 2006501043 A 1/2006
(Continued)

Primary Examiner — Kami A Bosworth

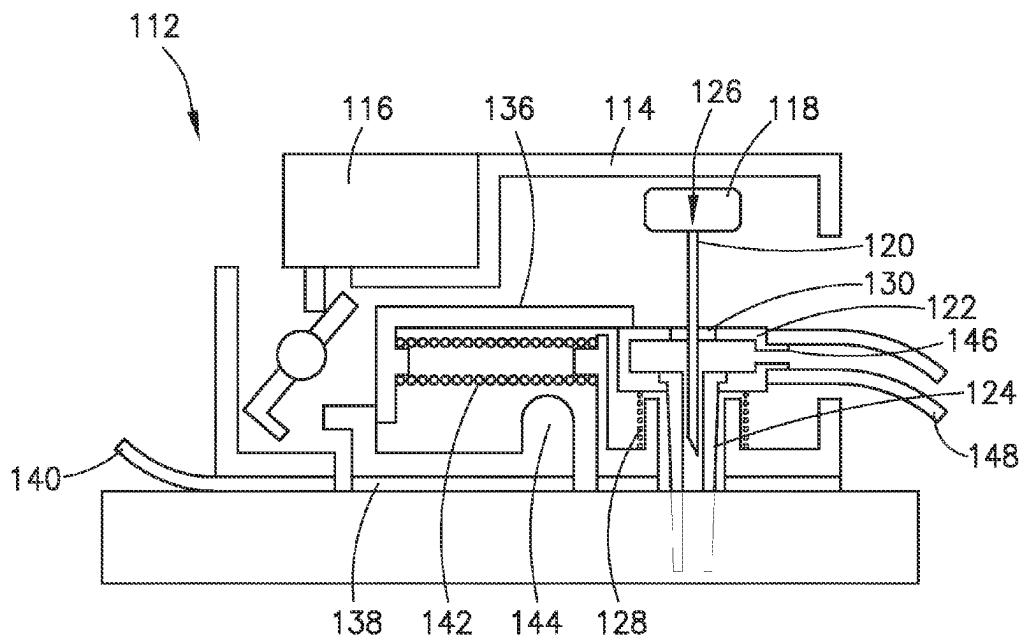
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(57) **ABSTRACT**

A needle hub for a drug delivery device includes a hub body, an activation button, a needle holder and a needle attached to the needle holder, a cannula holder and a cannula attached to the catheter holder, a needle actuation mechanism configured to move the needle holder and the cannula holder from a retracted position to an insertion position and configured to move the needle holder back to the retracted position, and a cannula spring biasing the catheter holder to a retracted position. Movement of the activation button is configured to cause the needle holder and the catheter holder to move from the retracted position to the insertion position, with the needle holder configured to return to the retracted position while the catheter holder remains in the insertion position.

5 Claims, 30 Drawing Sheets



- (52) **U.S. Cl.**
 CPC ... *A61M 5/162* (2013.01); *A61M 2005/14252*
 (2013.01); *A61M 2005/14256* (2013.01);
A61M 2005/1426 (2013.01); *A61M 2005/1585*
 (2013.01); *A61M 2005/1586* (2013.01); *A61M*
2005/1587 (2013.01)
- (58) **Field of Classification Search**
 CPC *A61M 5/3158*; *A61M 25/0102*; *A61M*
2005/14252; *A61M 2005/1585*; *A61M*
5/32933; *A61M 2005/1586*
 See application file for complete search history.
- (56) **References Cited**
 U.S. PATENT DOCUMENTS
- | | | | | | | | |
|--------------|-----|---------|--|--------------|-----|---------|--|
| 6,607,543 | B2 | 8/2003 | Purcell et al. | 2009/0054866 | A1 | 2/2009 | Teisen-Simony et al. |
| 7,250,037 | B2 | 7/2007 | Shermer et al. | 2010/0049129 | A1 | 2/2010 | Yokoi et al. |
| 7,931,621 | B2 | 4/2011 | Cross et al. | 2010/0100048 | A1 | 4/2010 | Nielsen et al. |
| 8,795,234 | B2 | 8/2014 | Kadamus et al. | 2011/0270220 | A1 | 11/2011 | Genosar |
| 9,466,757 | B2 | 10/2016 | Modi et al. | 2011/0270232 | A1 | 11/2011 | Forsell |
| 9,504,785 | B2 | 11/2016 | Forsell | 2011/0275996 | A1 | 11/2011 | Gyory et al. |
| 9,533,092 | B2 | 1/2017 | Gym | 2013/0012889 | A1 | 1/2013 | Schrage |
| 10,434,285 | B2 | 10/2019 | Schoonmaker et al. | 2013/0060233 | A1 | 3/2013 | O'Connor et al. |
| 10,933,191 | B2 | 3/2021 | Lawrence et al. | 2014/0058353 | A1* | 2/2014 | Politis A61M 5/14248
604/164.04 |
| 11,103,636 | B2 | 8/2021 | Olivas et al. | 2014/0088509 | A1 | 3/2014 | Sonderegger et al. |
| 2002/0123740 | A1 | 9/2002 | Flaherty et al. | 2014/0316379 | A1 | 10/2014 | Sonderegger et al. |
| 2004/0158207 | A1 | 8/2004 | Hunn et al. | 2015/0005716 | A1 | 1/2015 | Adair et al. |
| 2005/0273019 | A1 | 12/2005 | Conway et al. | 2016/0058941 | A1 | 3/2016 | Wu et al. |
| 2006/0161108 | A1 | 7/2006 | Mogensen et al. | 2016/0296695 | A1 | 10/2016 | Hassman et al. |
| 2008/0051738 | A1 | 2/2008 | Griffin | 2017/0021137 | A1 | 1/2017 | Cole |
| 2008/0058719 | A1 | 3/2008 | Edwards et al. | 2017/0043133 | A1 | 2/2017 | Amano et al. |
| 2008/0195049 | A1 | 8/2008 | Thalmann et al. | 2017/0259011 | A1 | 9/2017 | Nielsen |
| 2008/0215006 | A1* | 9/2008 | Thorkild A61M 25/0606
604/151 | 2018/0021508 | A1 | 1/2018 | Destefano et al. |
| 2009/0012472 | A1 | 1/2009 | Ahm et al. | 2019/0015584 | A1 | 1/2019 | Meehan et al. |
| | | | | 2019/0053465 | A1* | 2/2019 | Knight A61D 7/00 |
| | | | | 2019/0060562 | A1 | 2/2019 | Olivas et al. |
| | | | | 2019/0175840 | A1 | 6/2019 | Schabbach et al. |
| | | | | 2019/0365986 | A1 | 12/2019 | Coiner et al. |
| | | | | 2019/0365993 | A1 | 12/2019 | Staub et al. |
| | | | | 2019/0388615 | A1 | 12/2019 | Sonderegger et al. |
| | | | | 2020/0001005 | A1 | 1/2020 | Politis et al. |
| | | | | 2020/0164155 | A1 | 5/2020 | Mojarrad et al. |
| | | | | 2020/0238004 | A1 | 7/2020 | Mccullough |
| | | | | 2020/0360235 | A1 | 11/2020 | Moller |
- FOREIGN PATENT DOCUMENTS
- | | | | |
|----|------------|----|--------|
| JP | 2020505175 | A | 2/2020 |
| WO | 0025844 | A1 | 5/2000 |
| WO | 2004032990 | A2 | 4/2004 |
- * cited by examiner

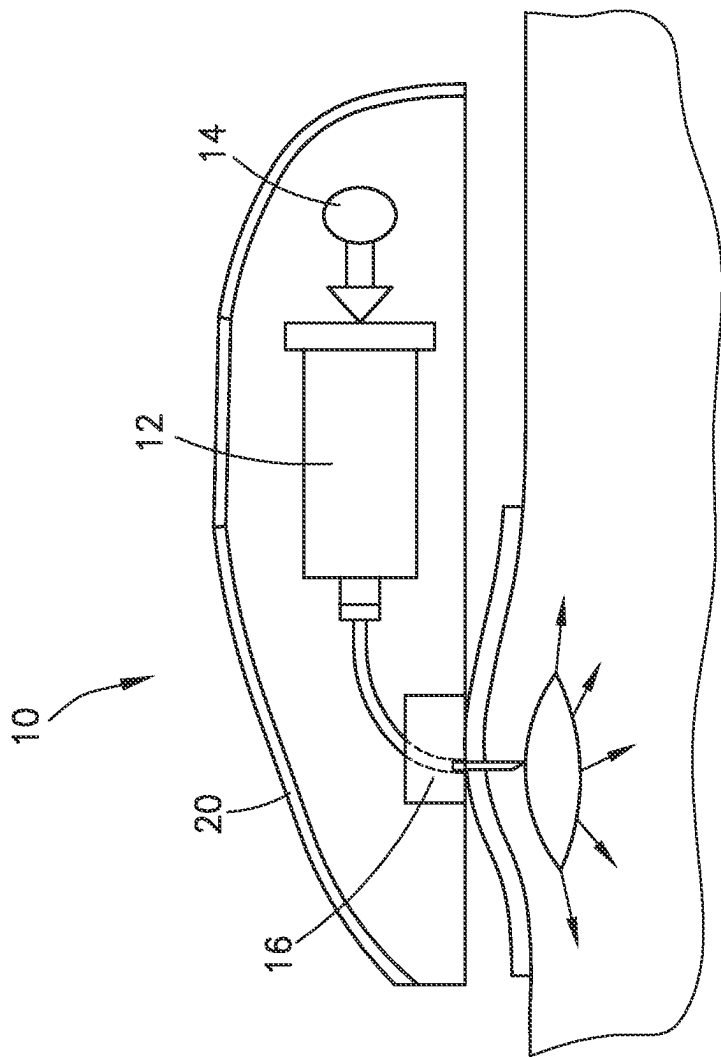
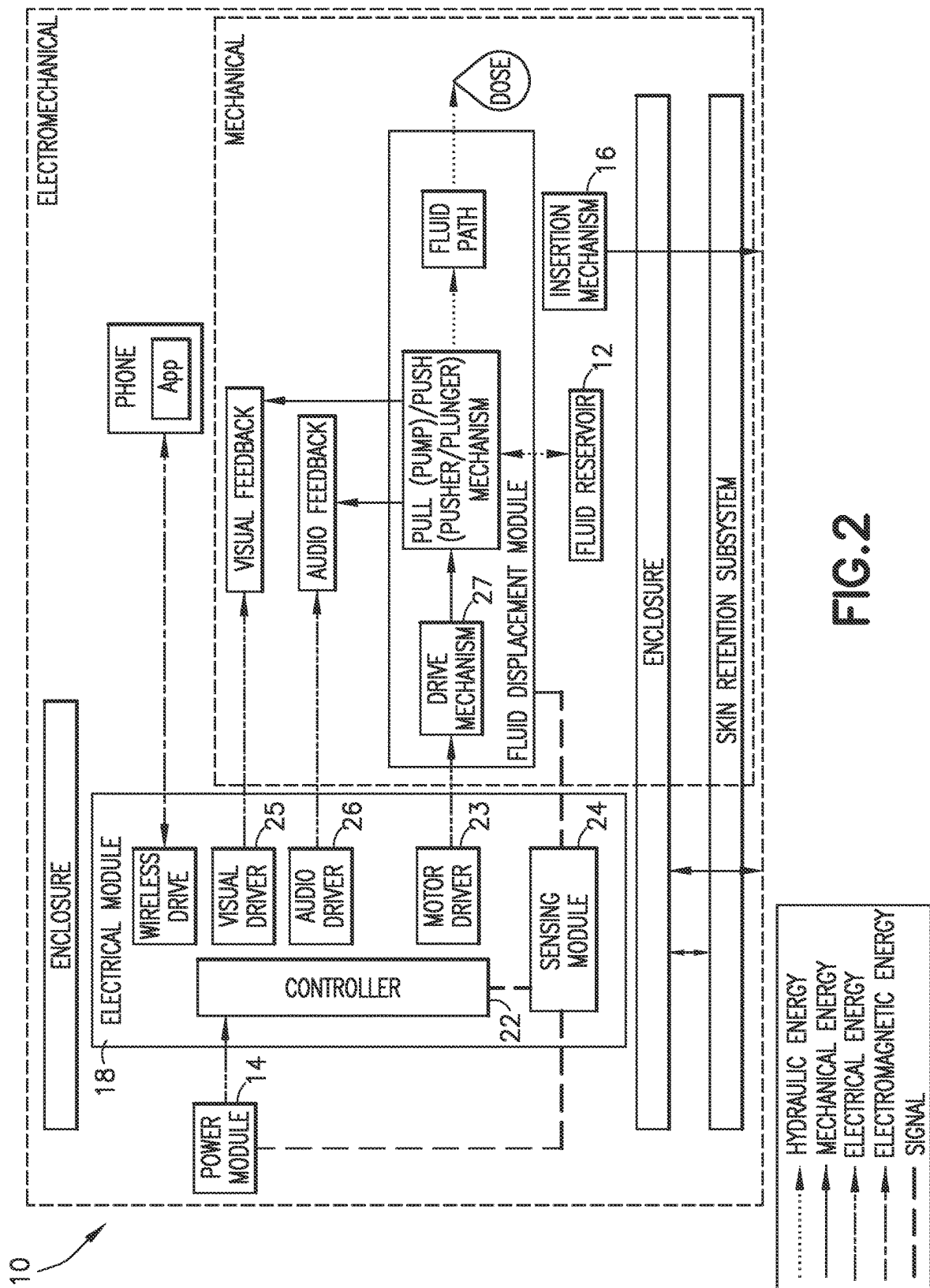
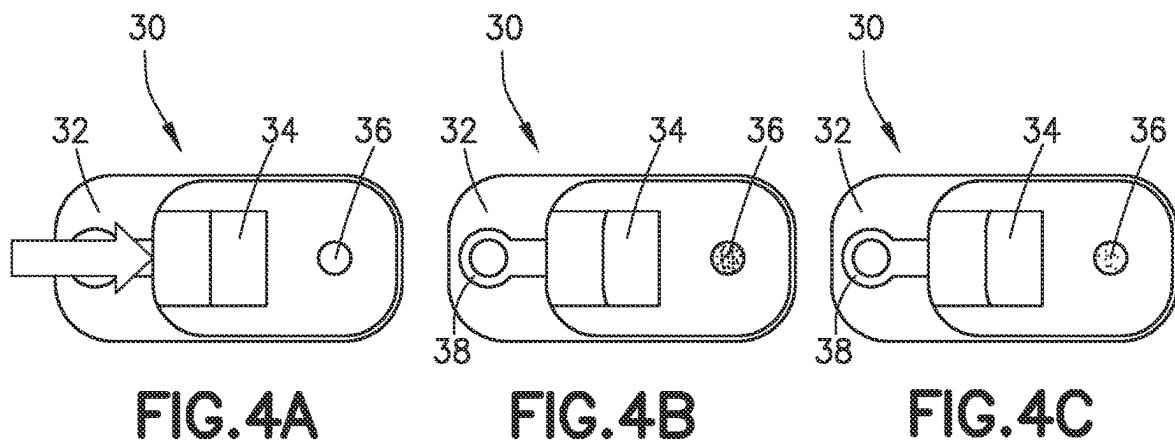
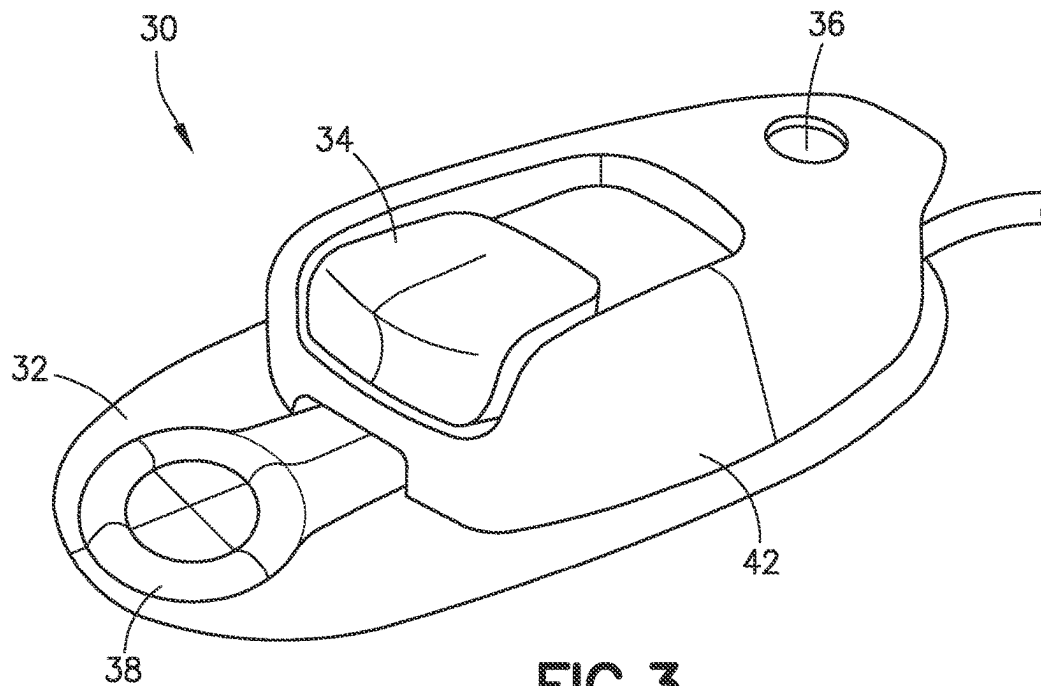
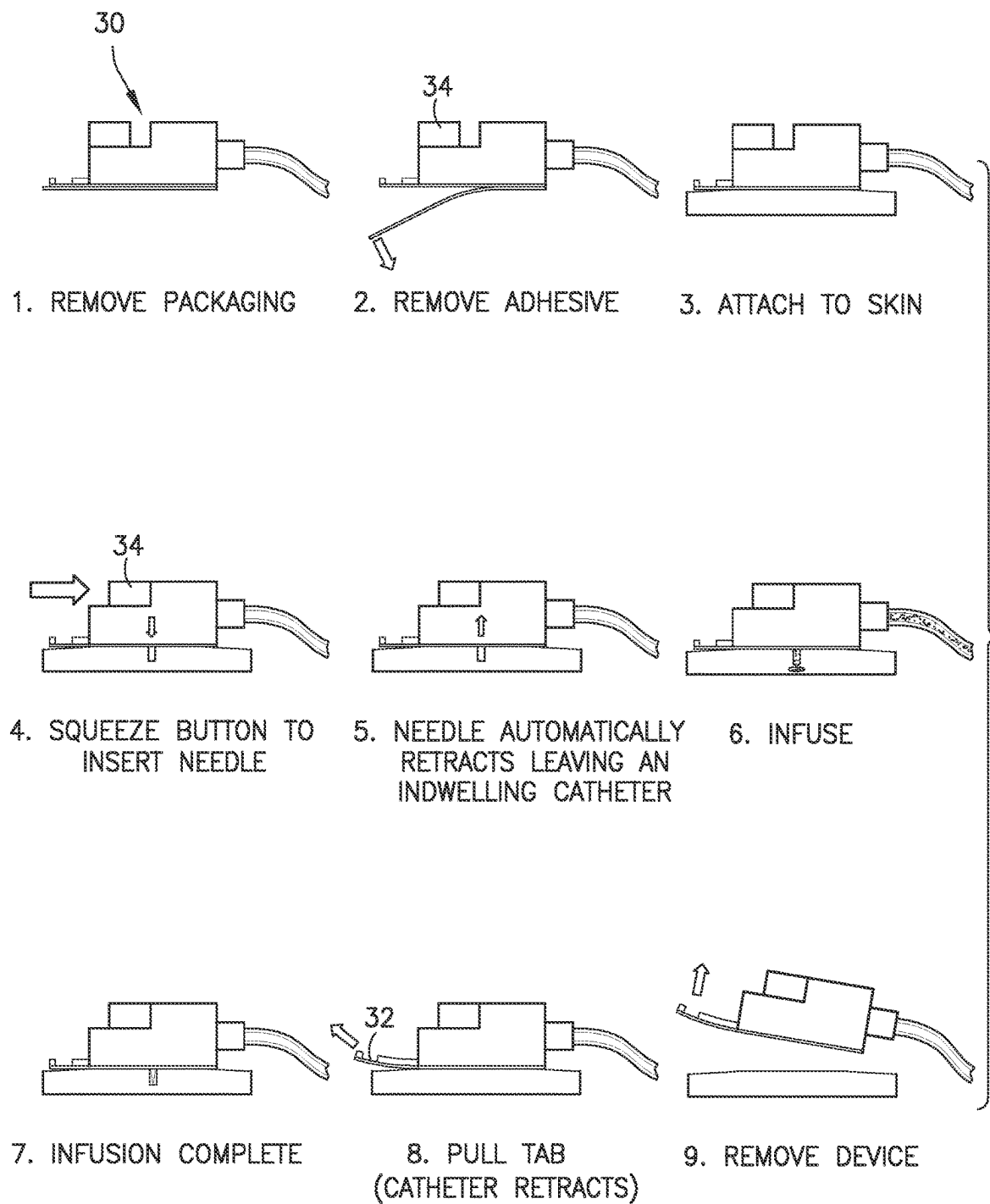


FIG.1





**FIG.5**

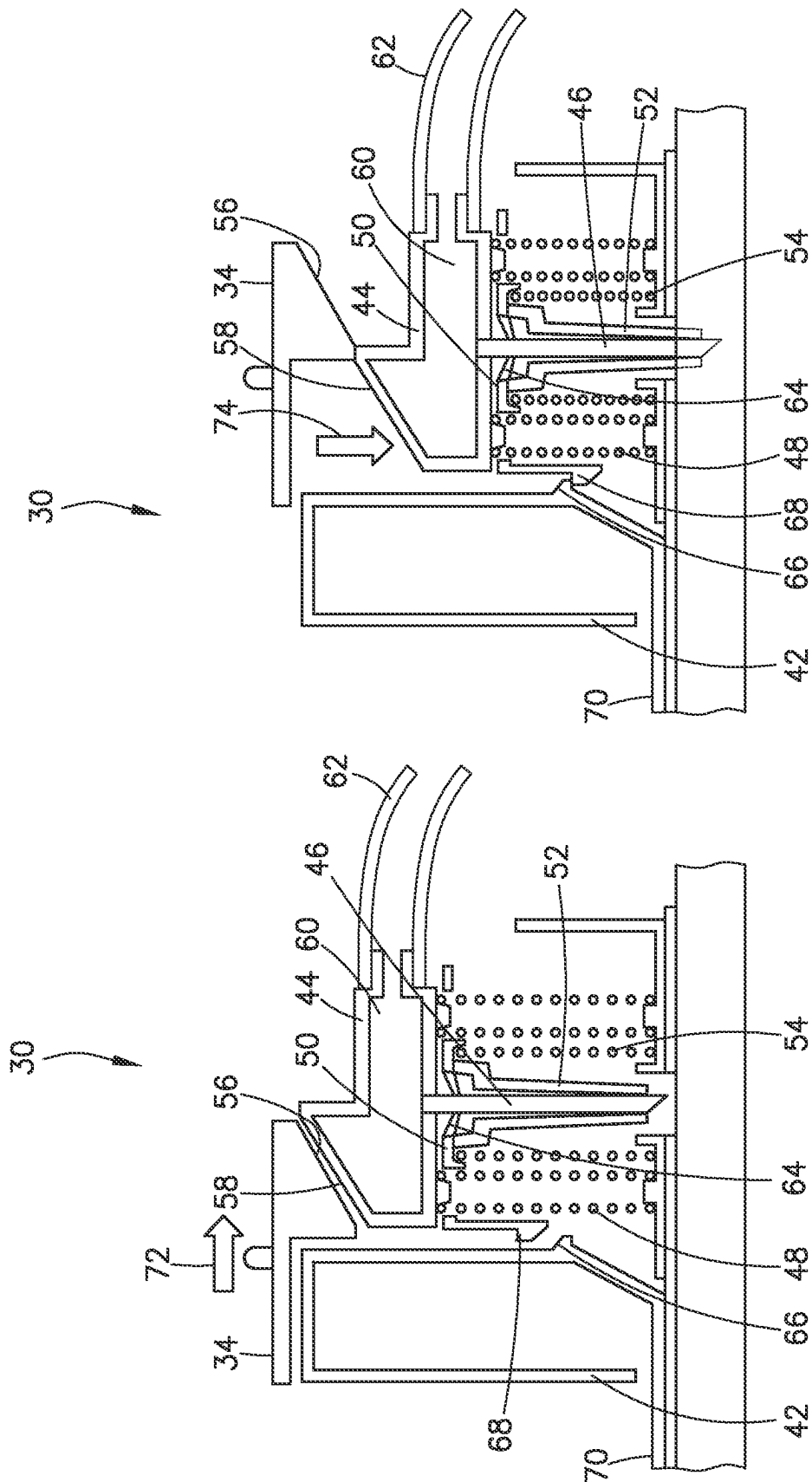
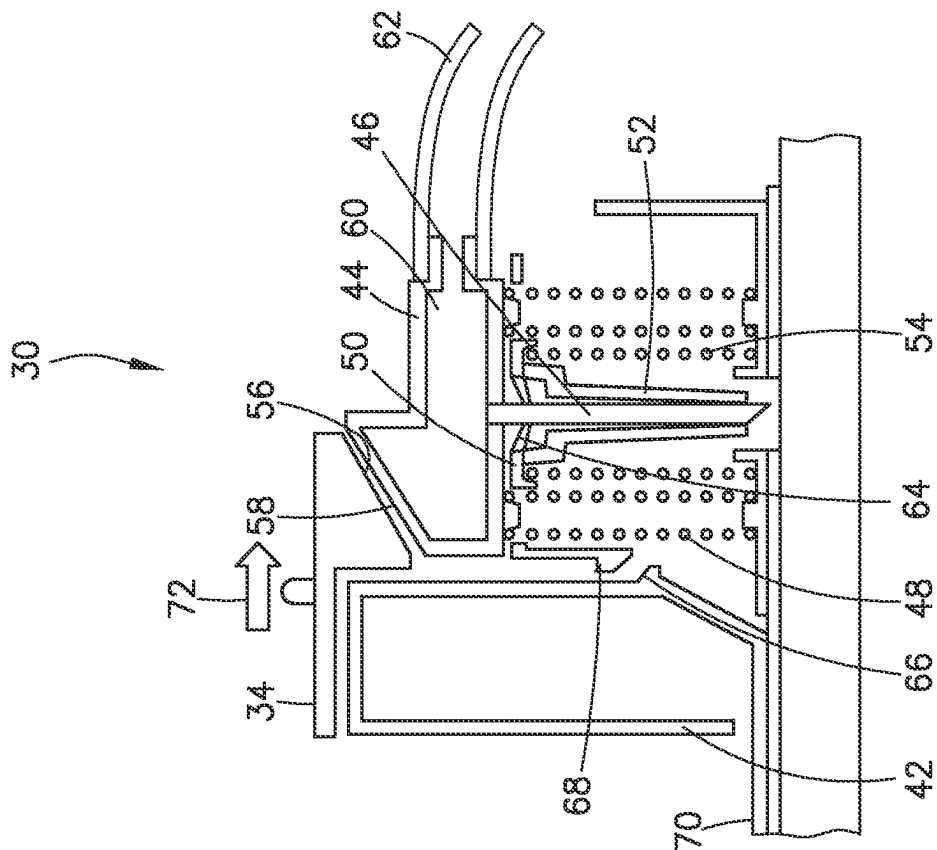
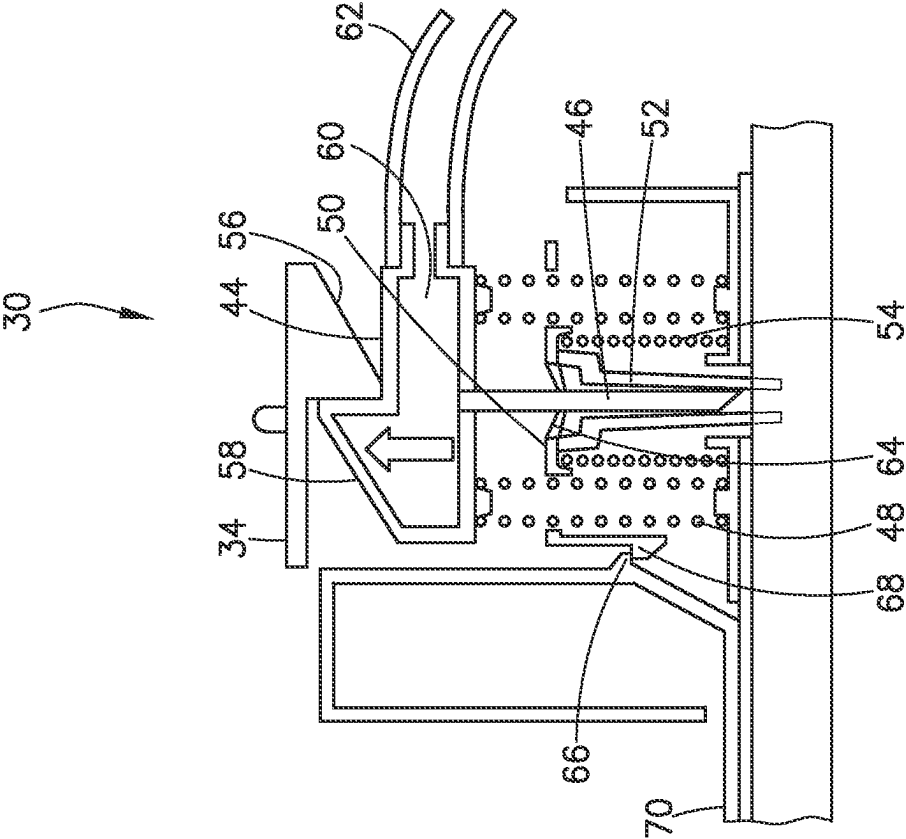
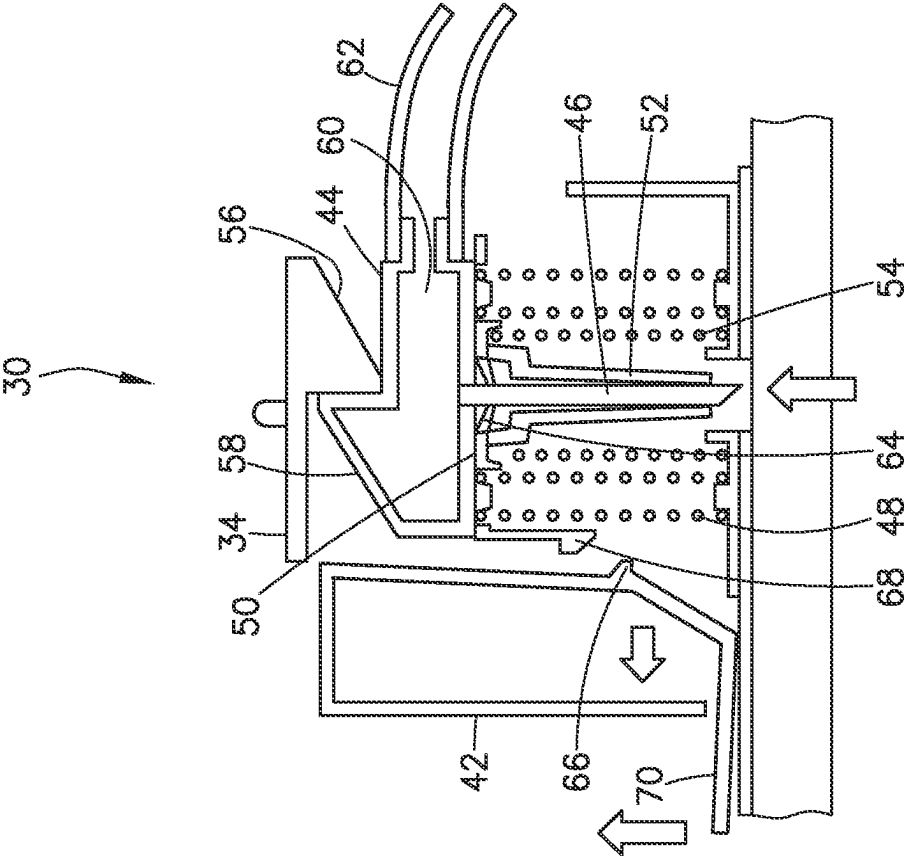
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FIG. 6A



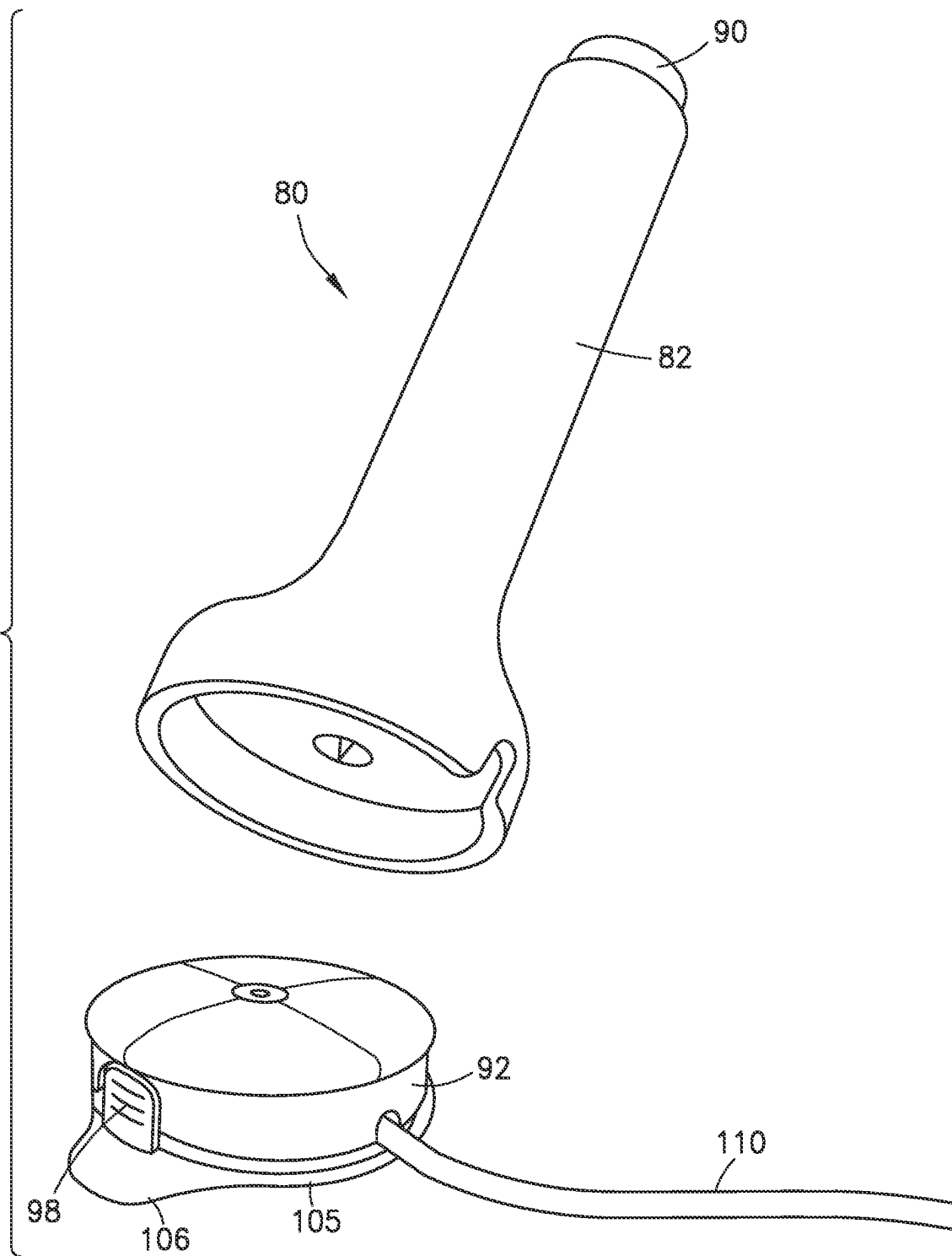


FIG. 7

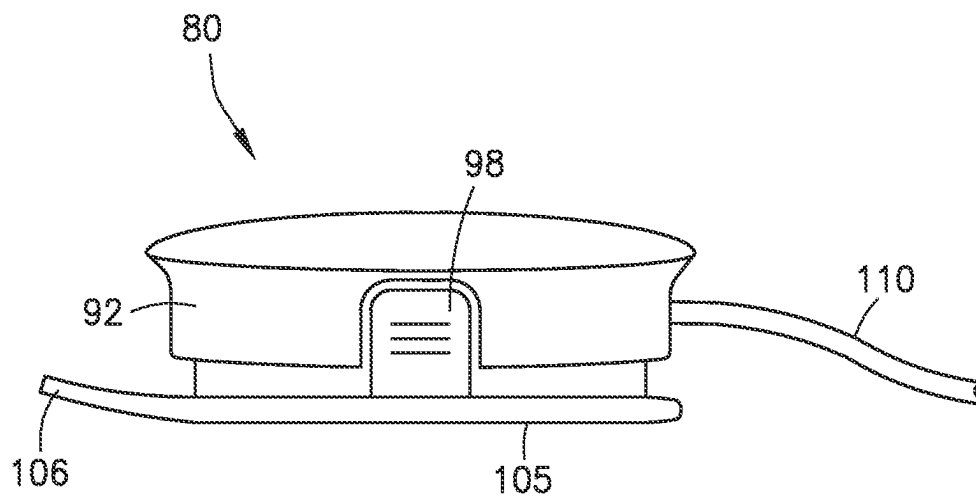


FIG. 8

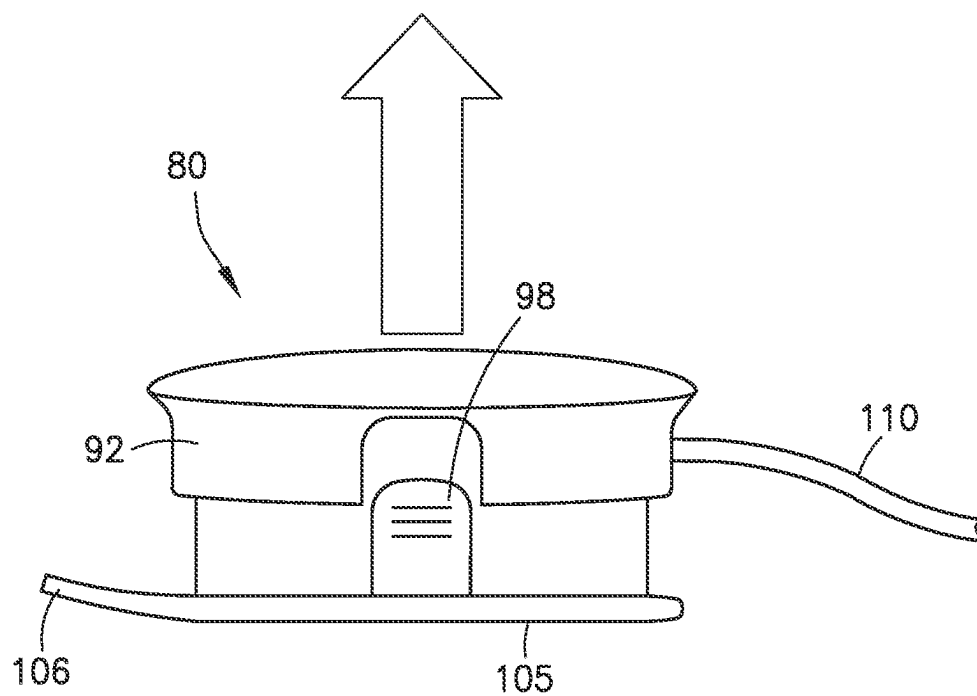


FIG. 9

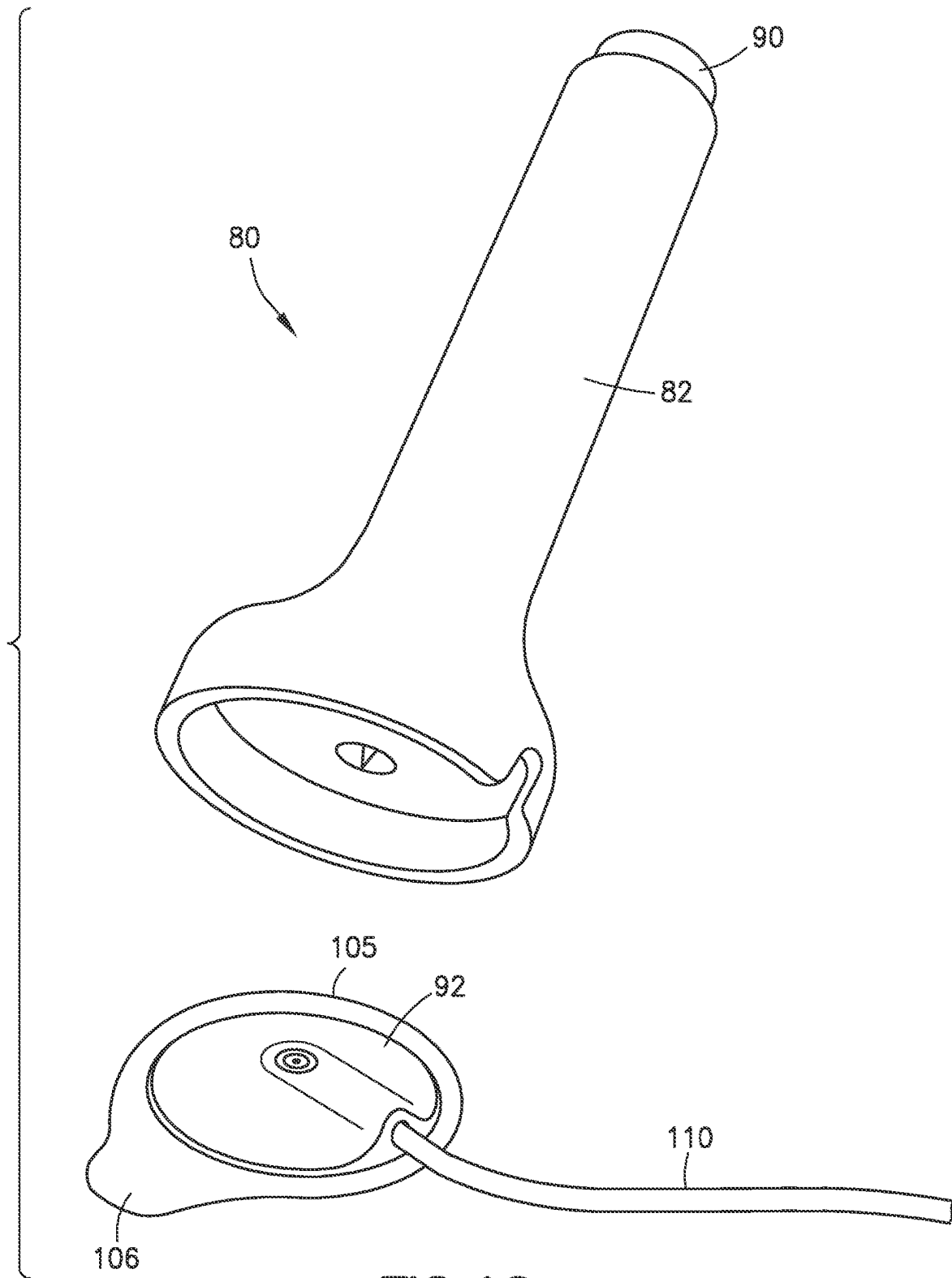


FIG.10

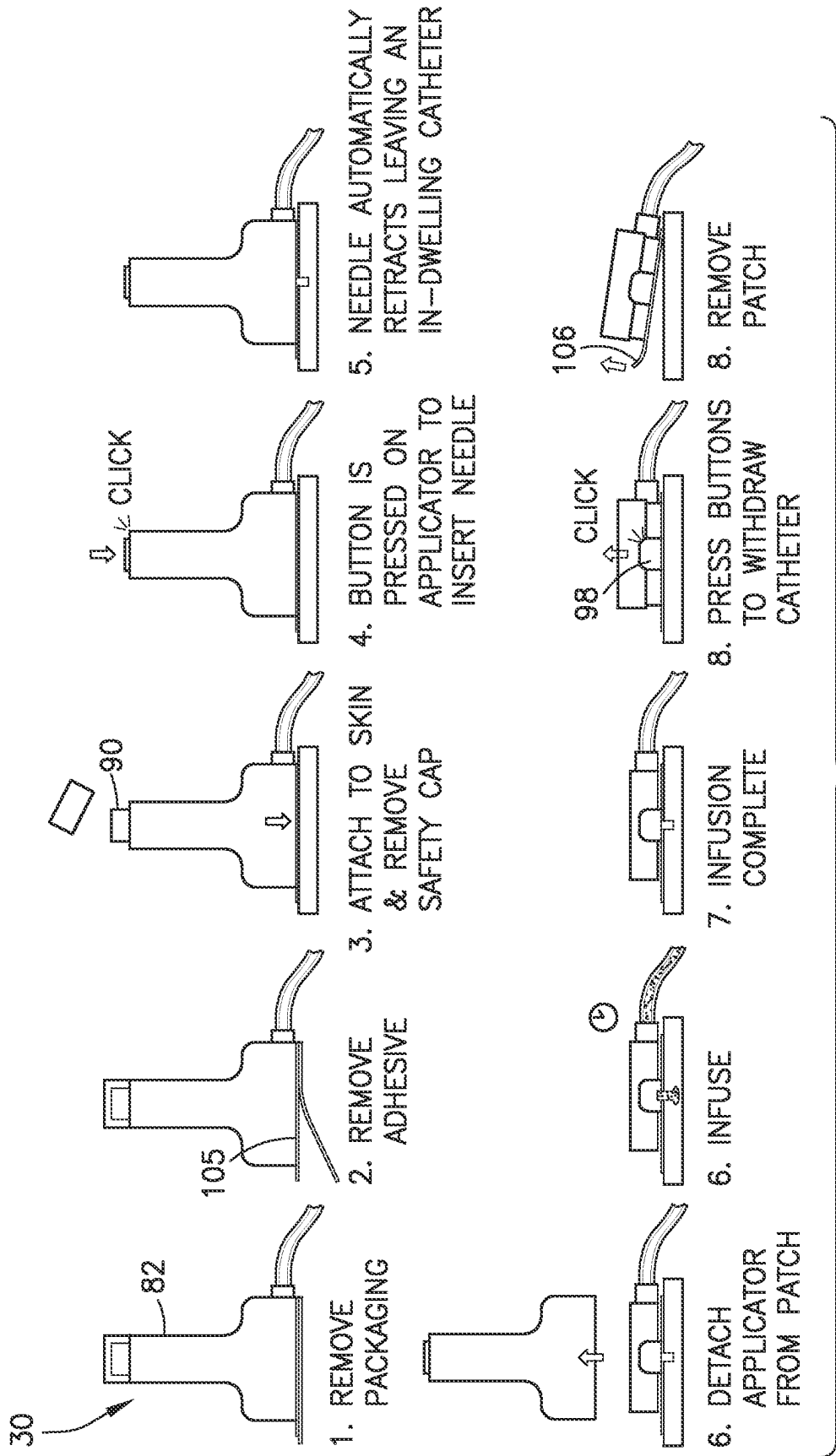


FIG.11

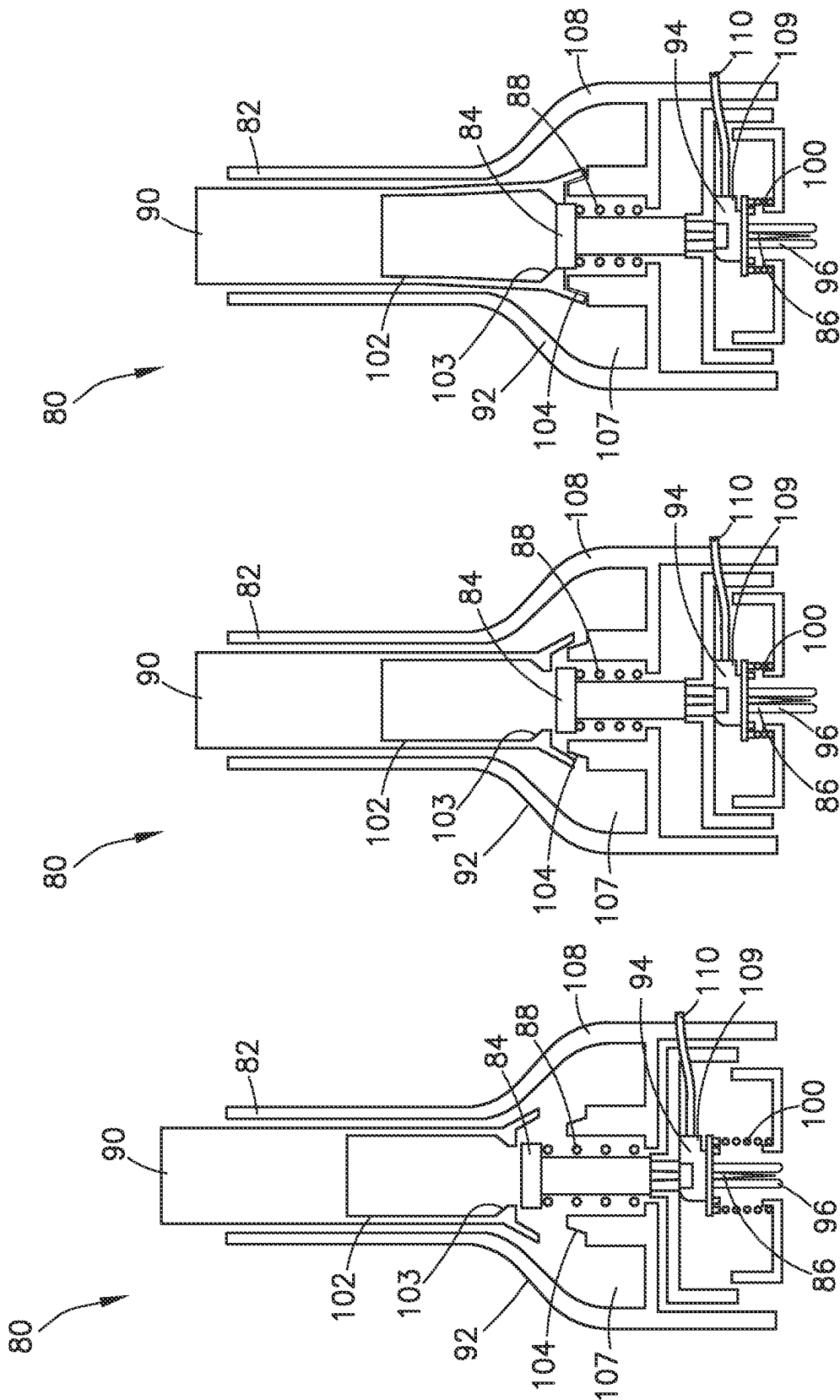


FIG. 12C

FIG. 12B

FIG. 12A

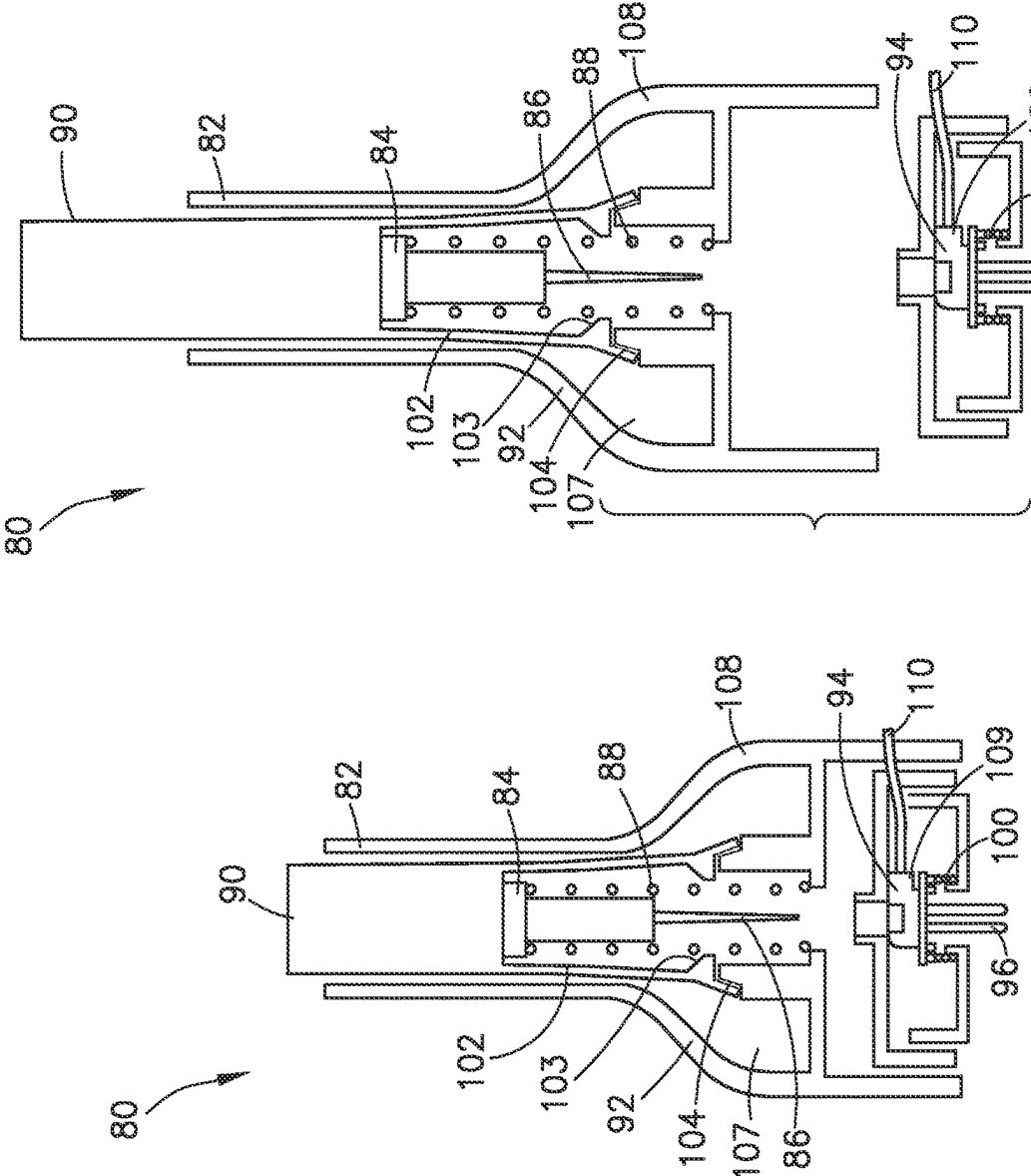


FIG. 12D

FIG. 12E

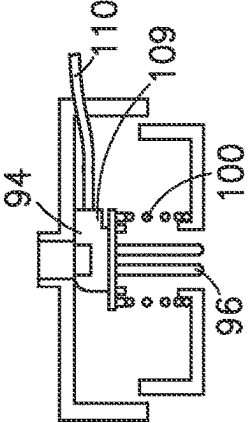


FIG. 12F

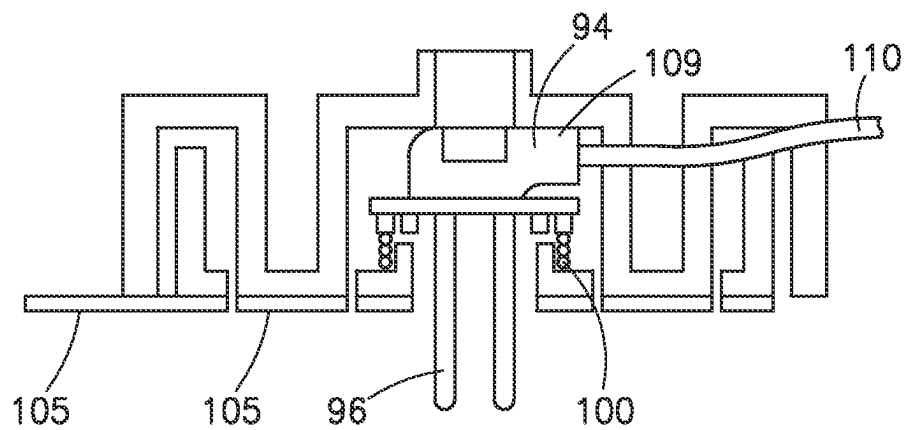


FIG. 13

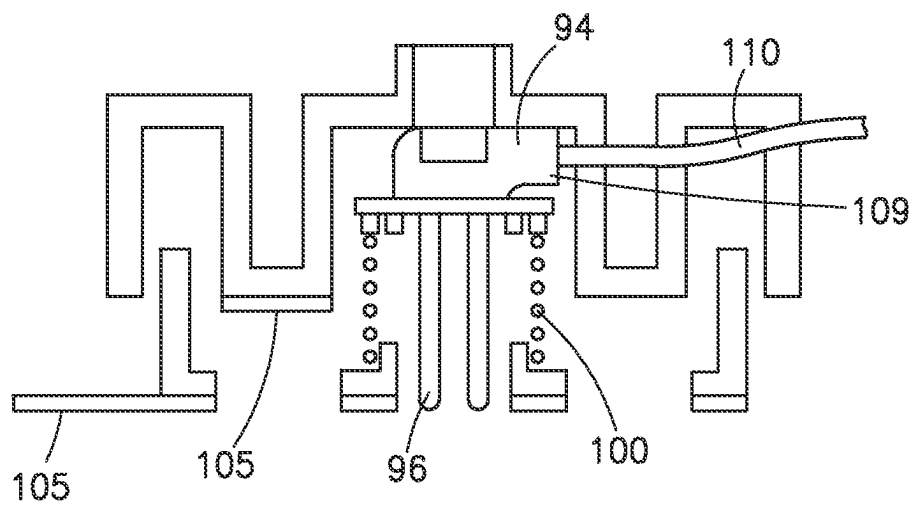
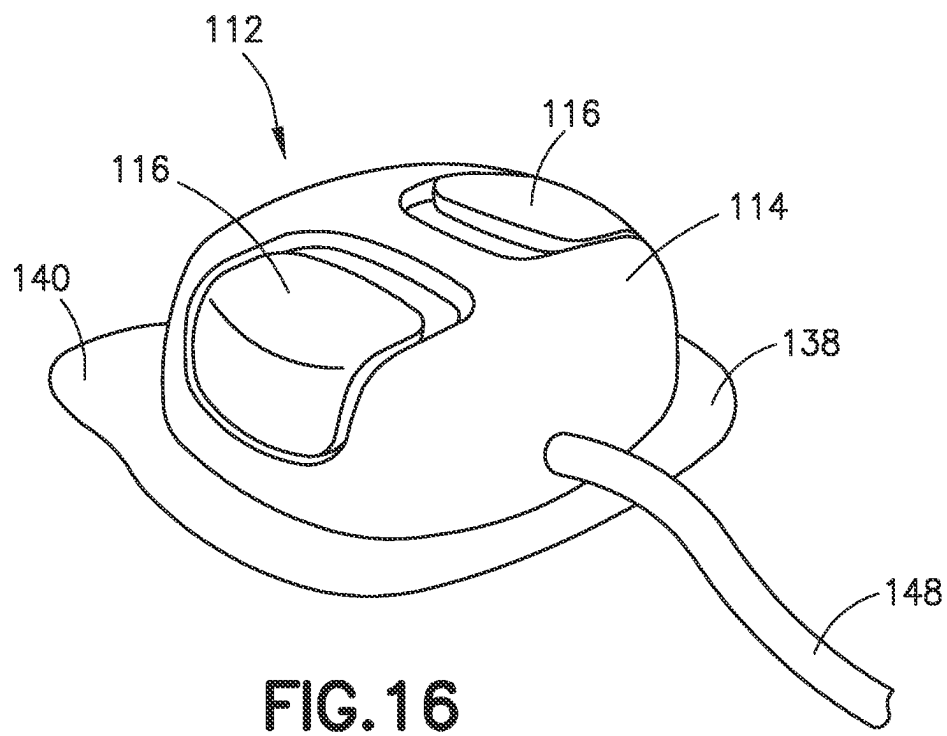
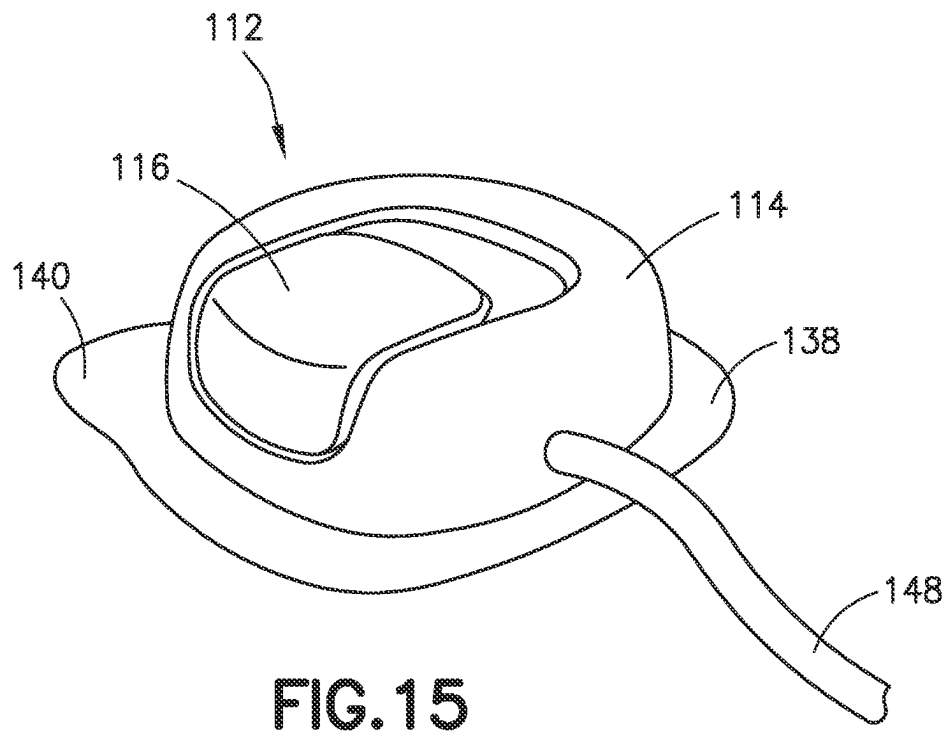


FIG. 14



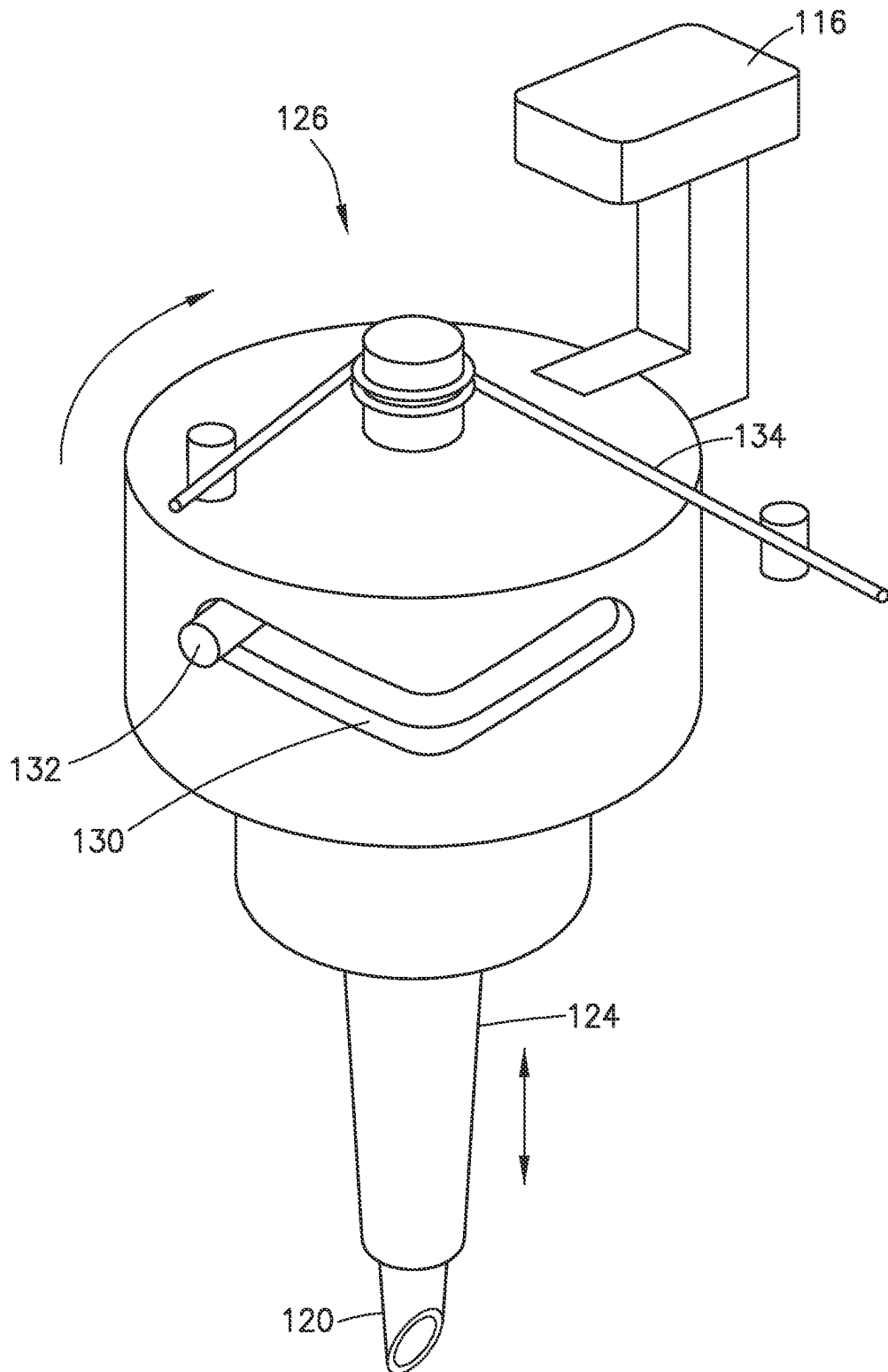


FIG.17

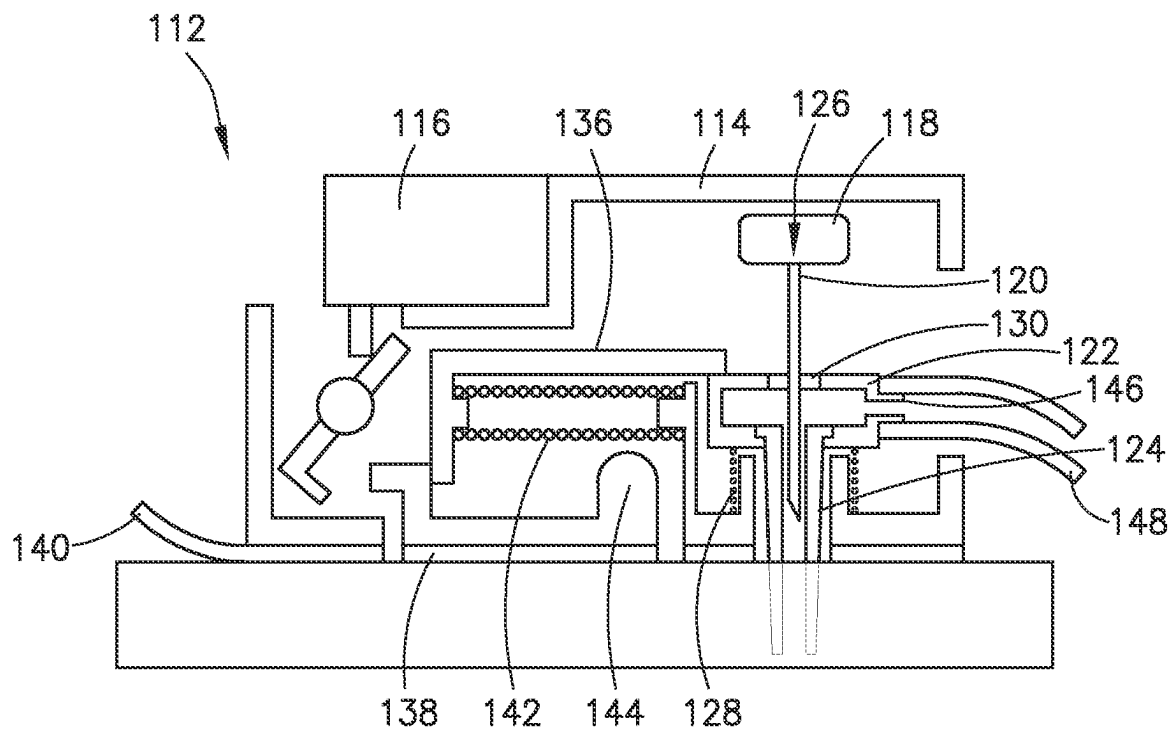


FIG. 18A

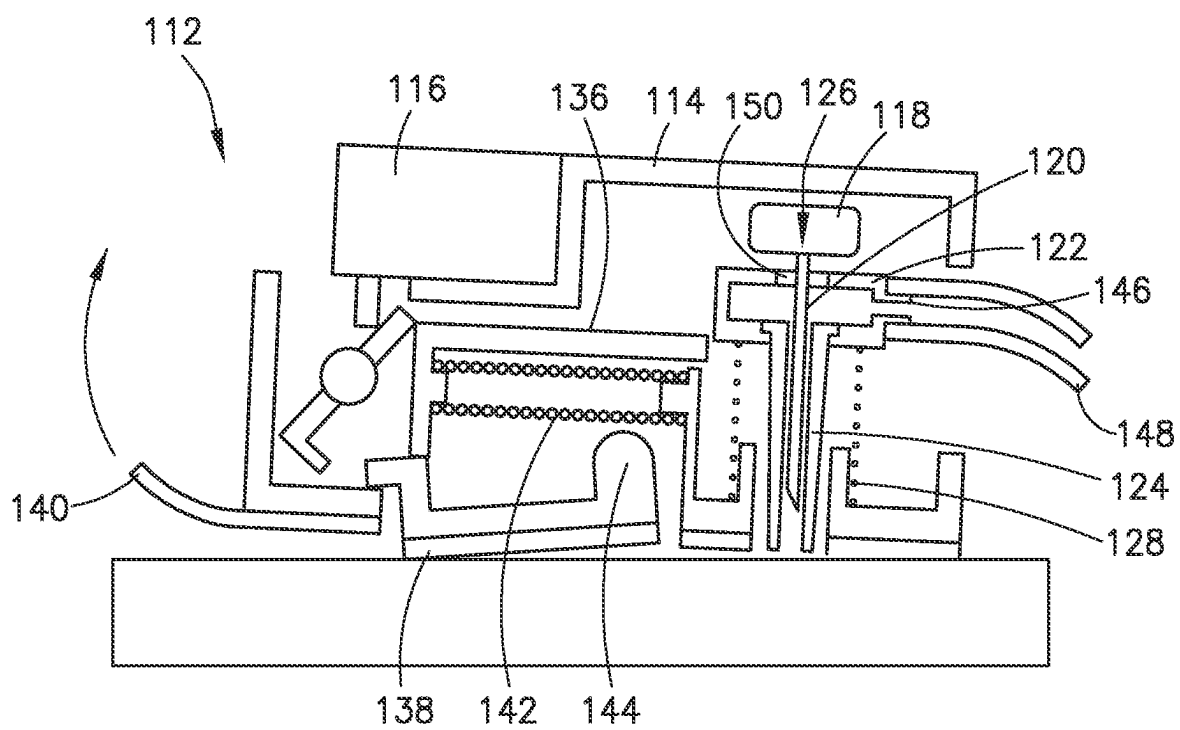


FIG. 18B

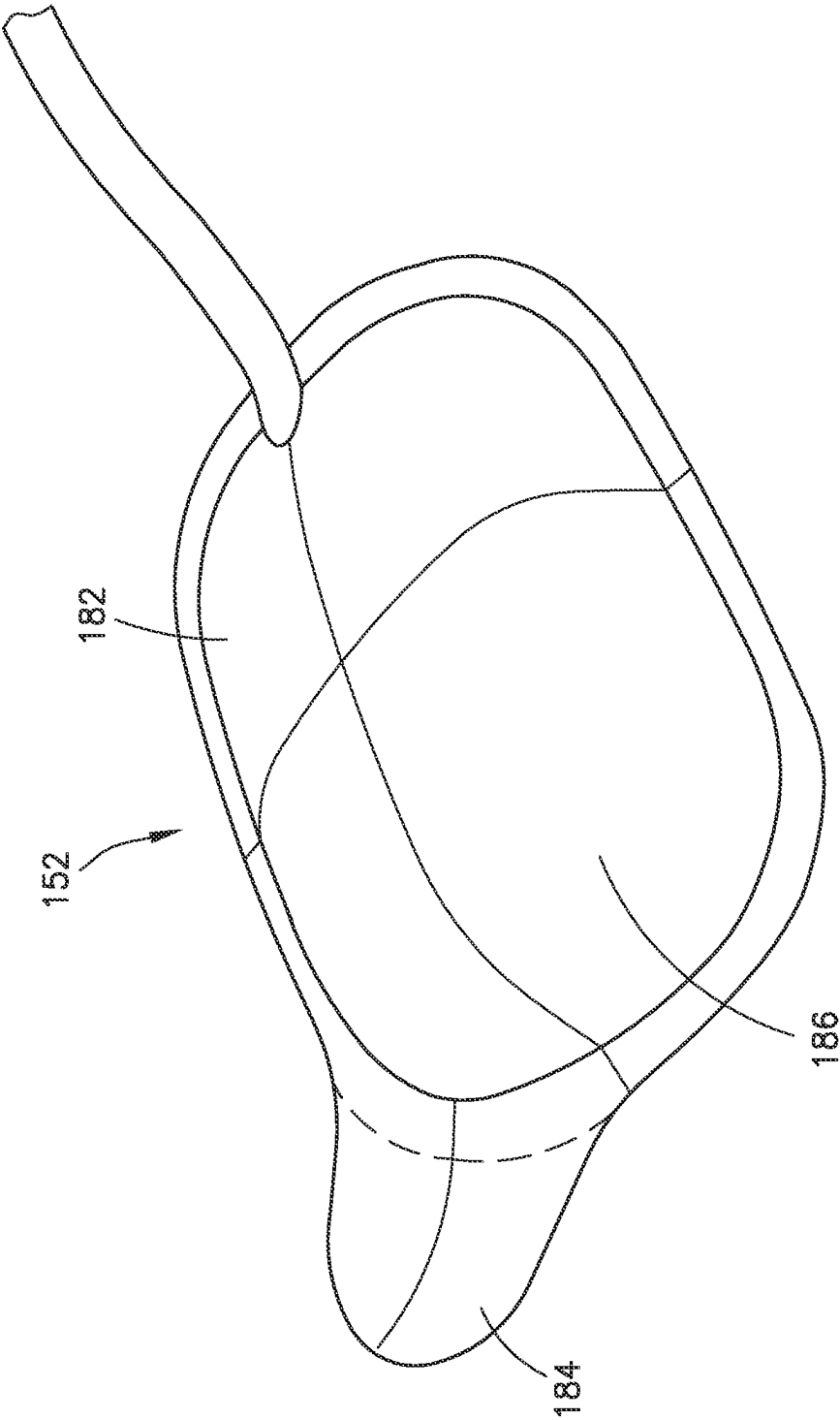


FIG.19

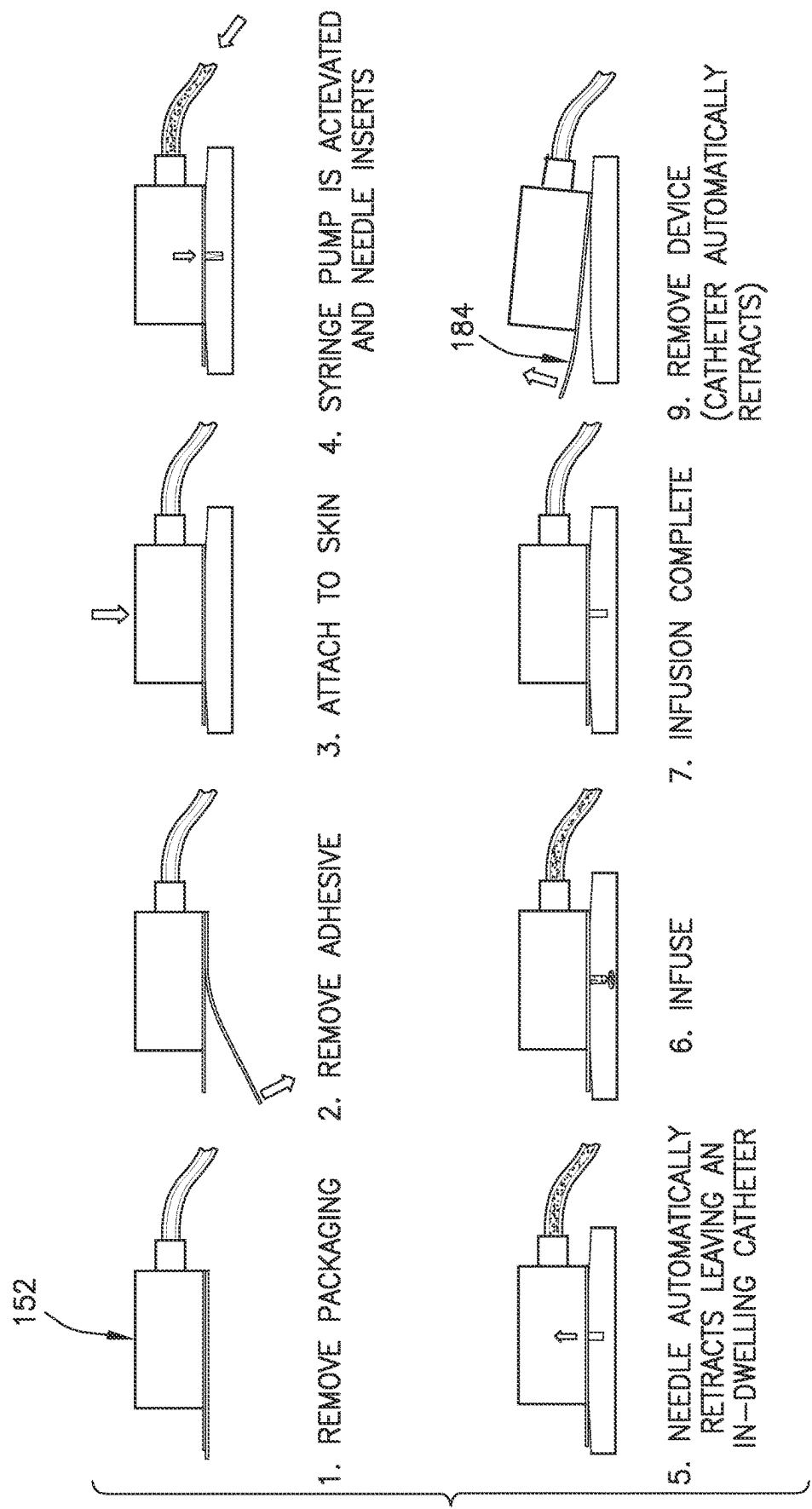


FIG.20

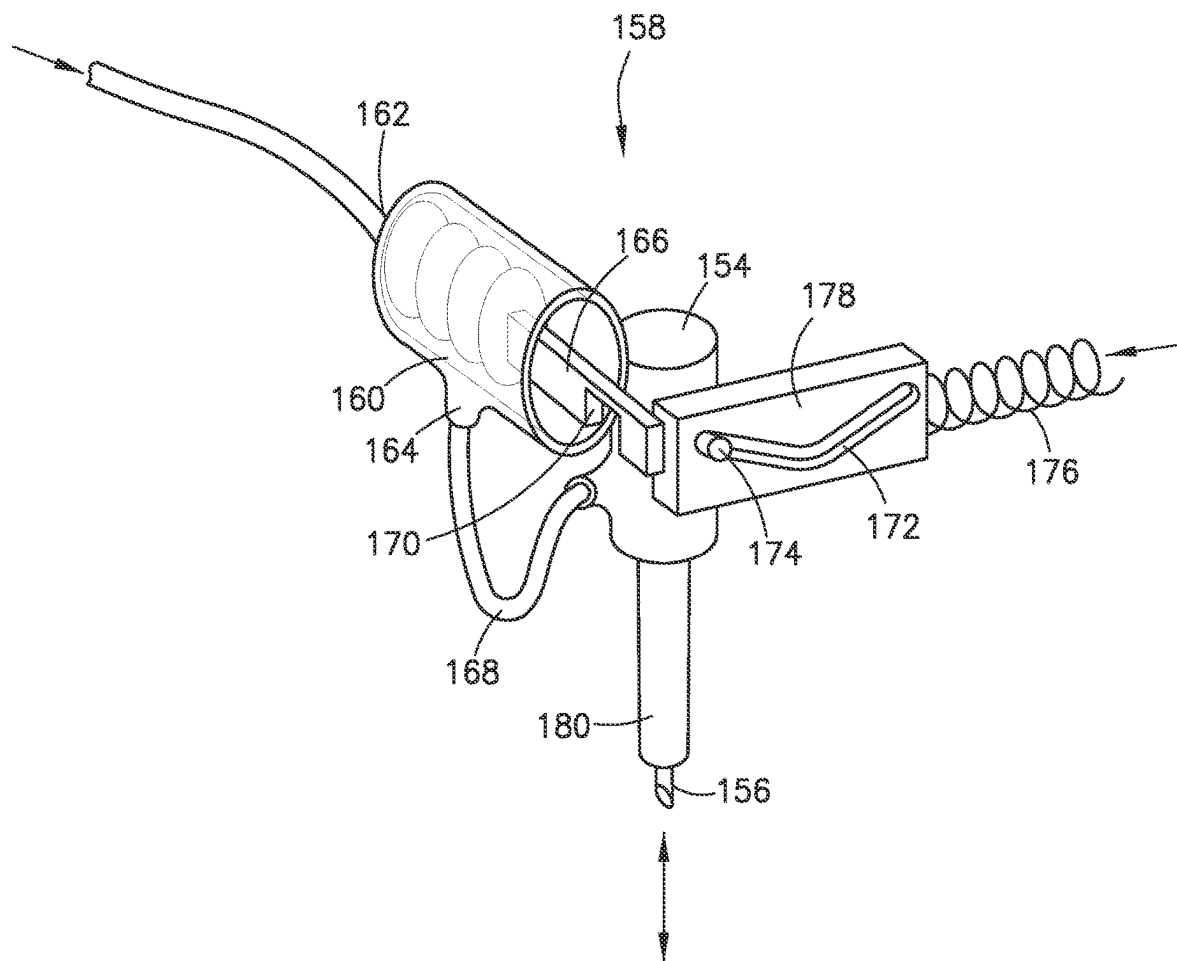


FIG. 21

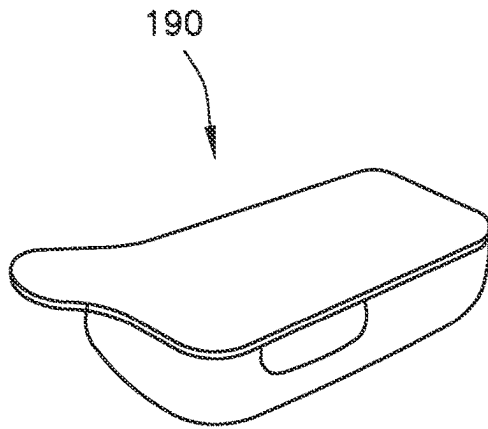


FIG. 22

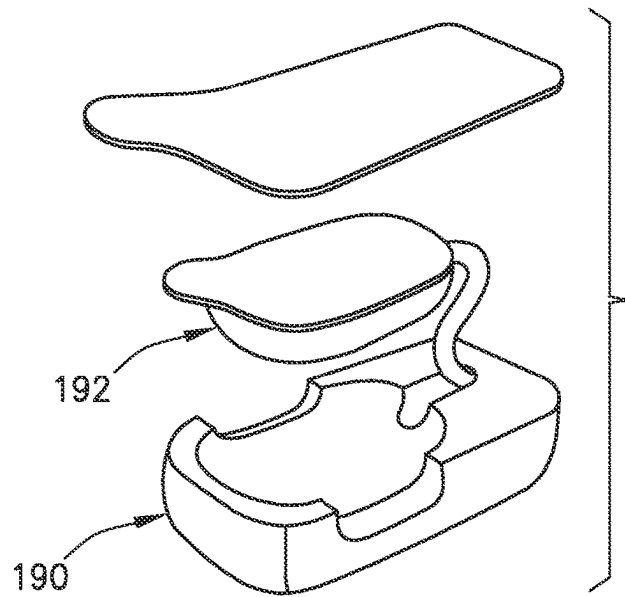


FIG. 23

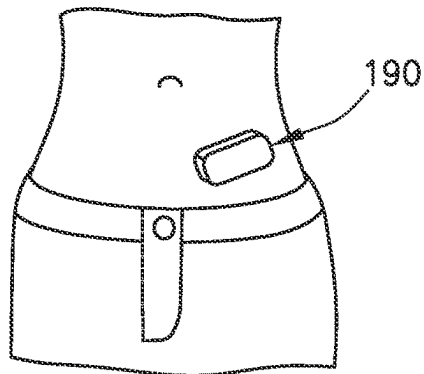


FIG. 24

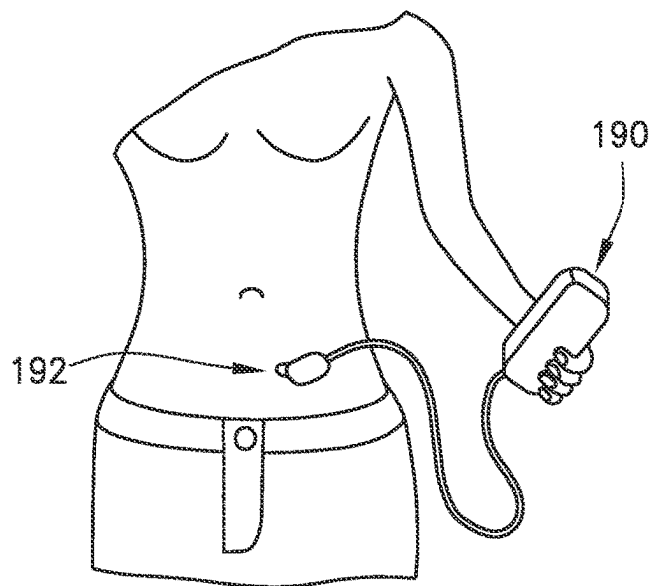


FIG. 25

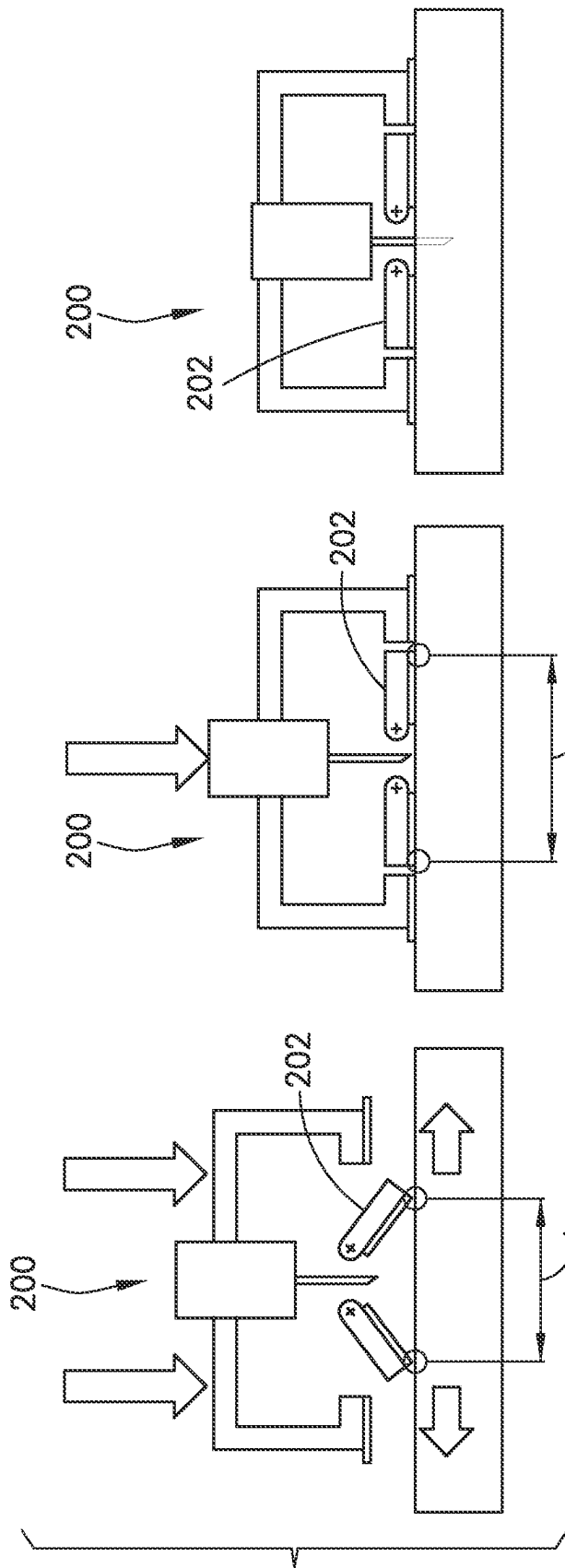


FIG. 26C

FIG. 26B

FIG. 26A

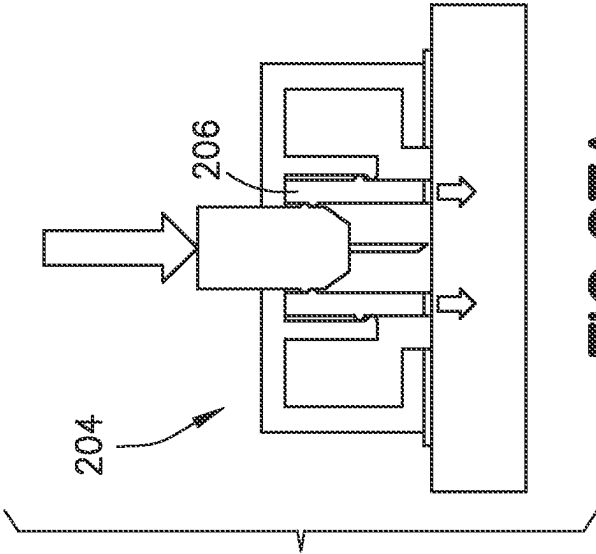


FIG. 27A

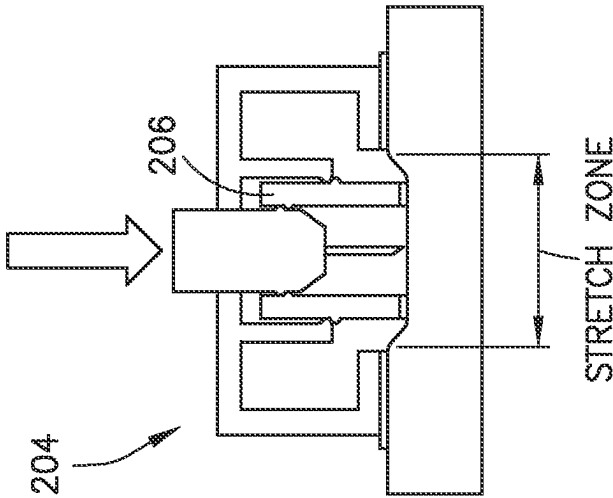


FIG. 27B

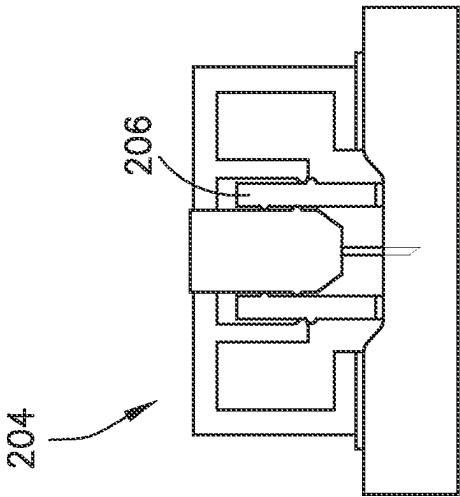


FIG. 27C

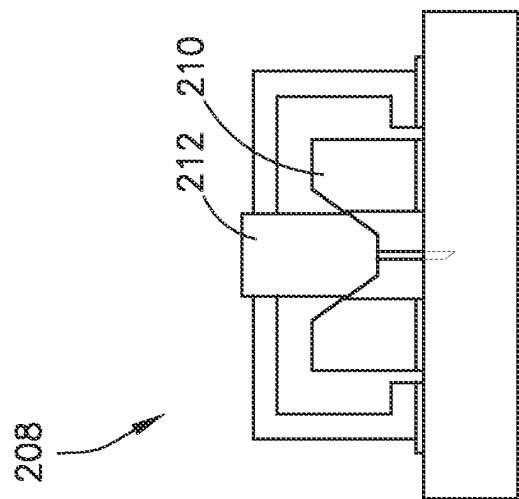


FIG. 28A

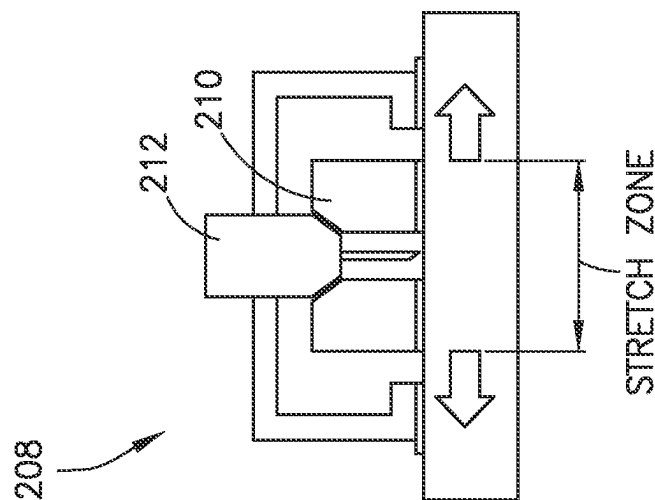


FIG. 28B

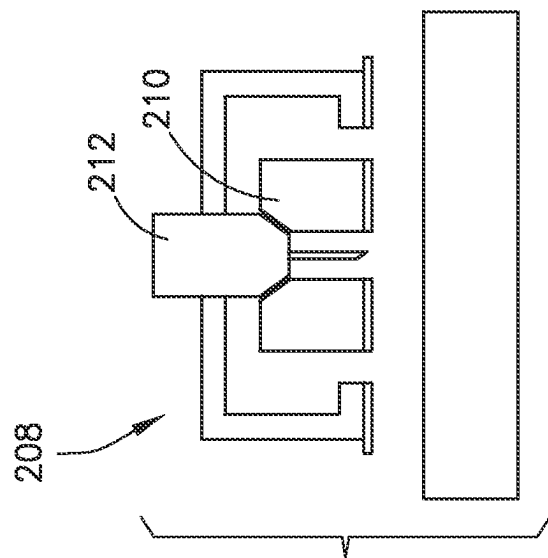


FIG. 28C

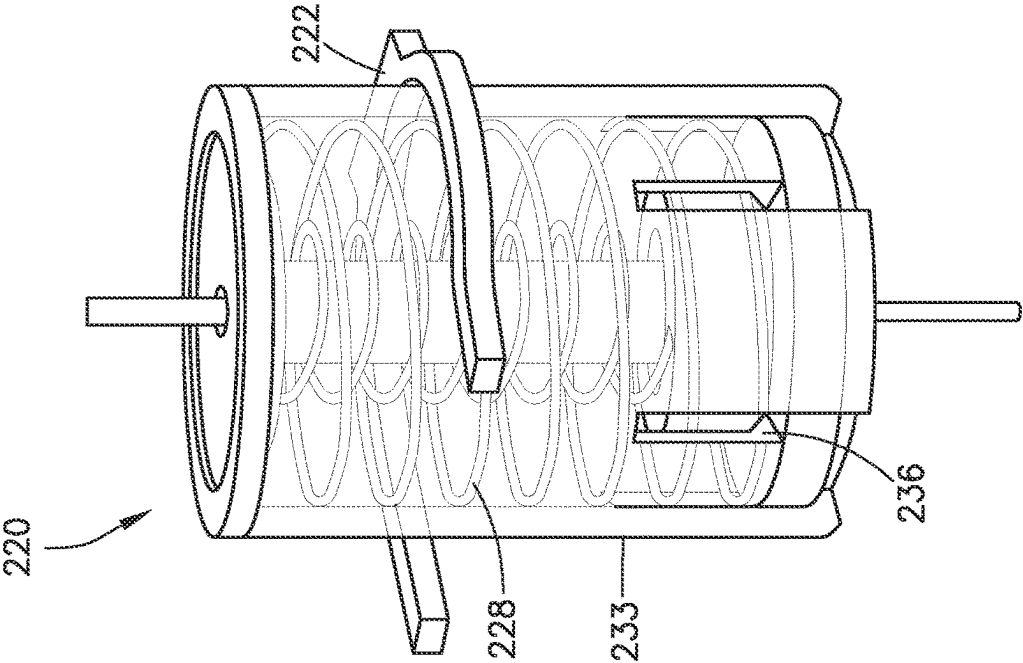


FIG.29

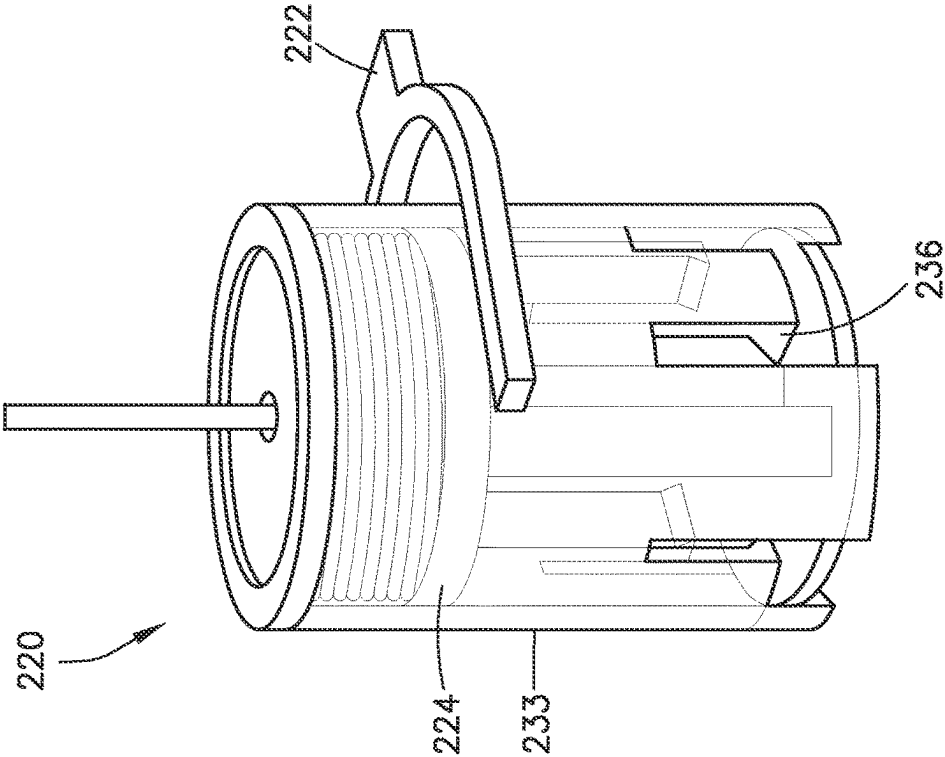


FIG.30

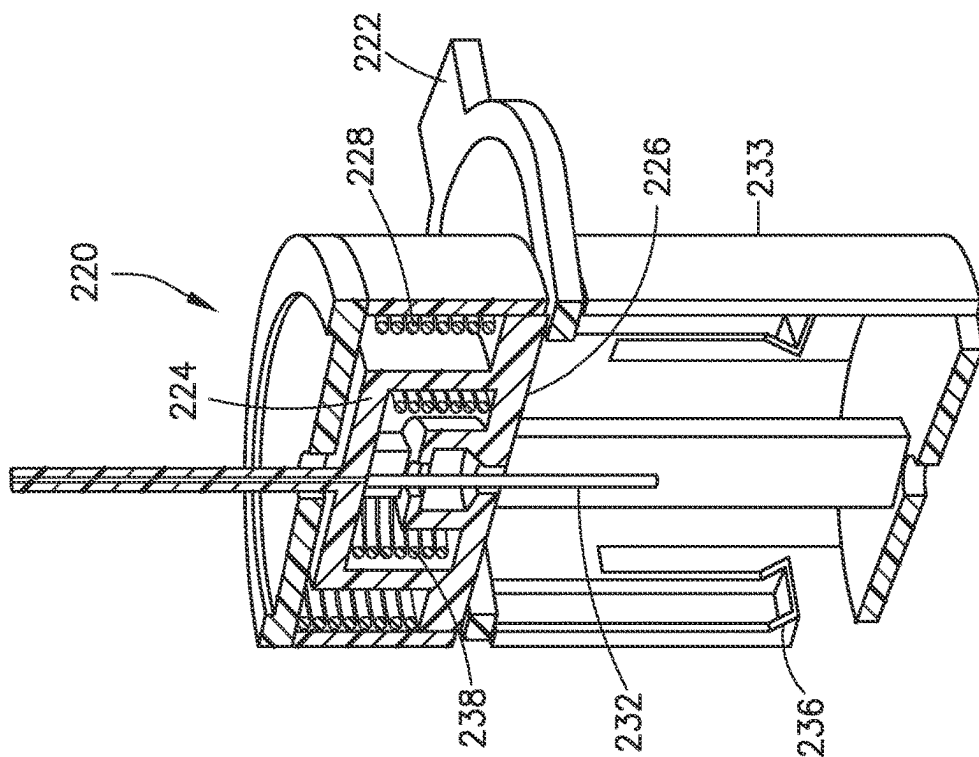


FIG. 31

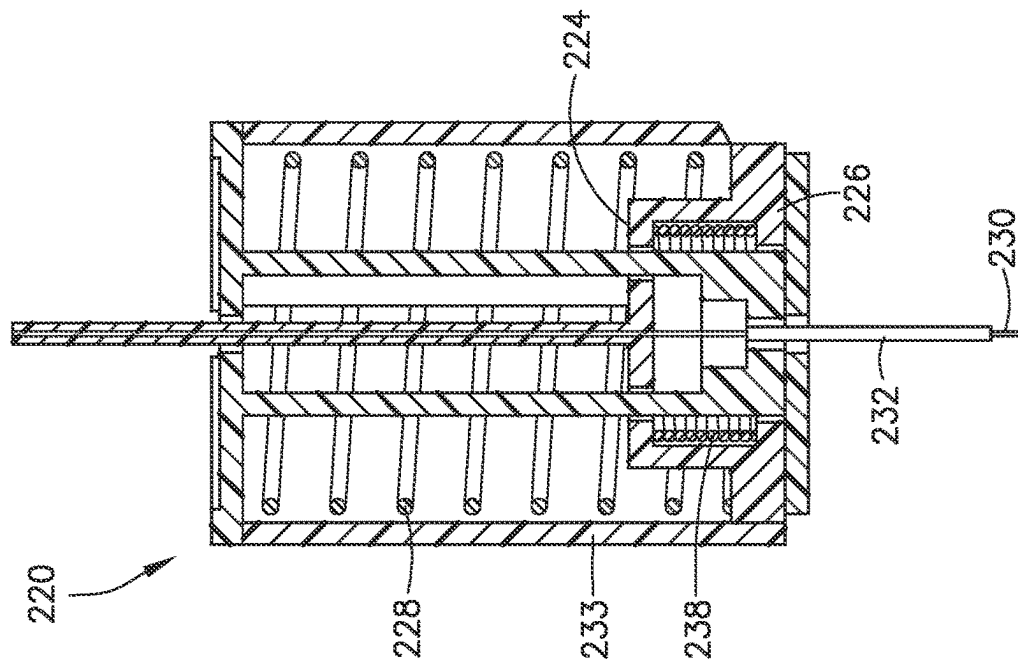


FIG. 32

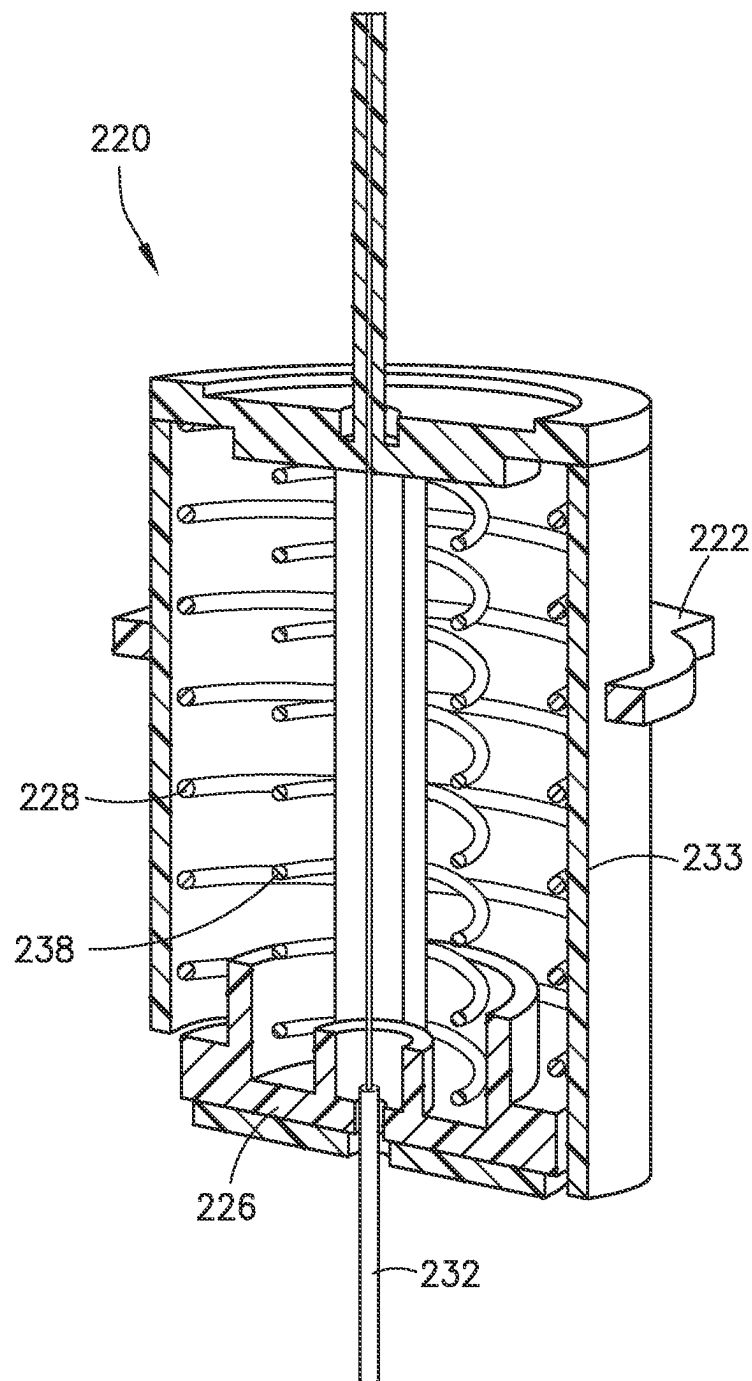


FIG. 33

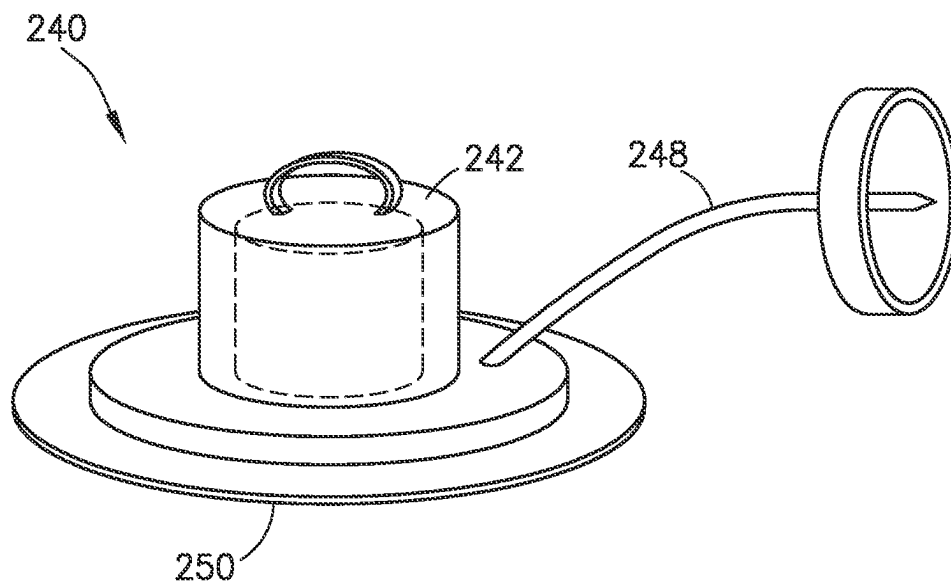


FIG. 34

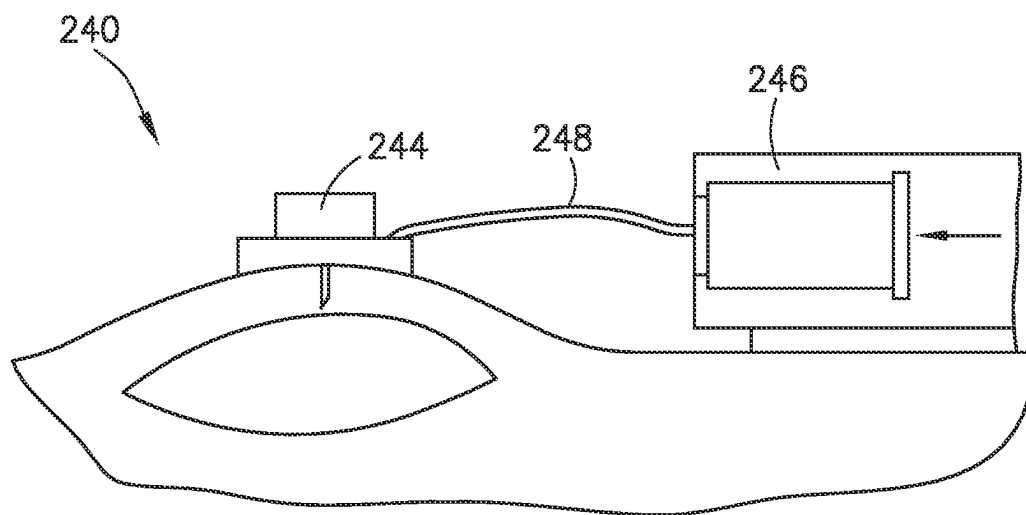


FIG. 35

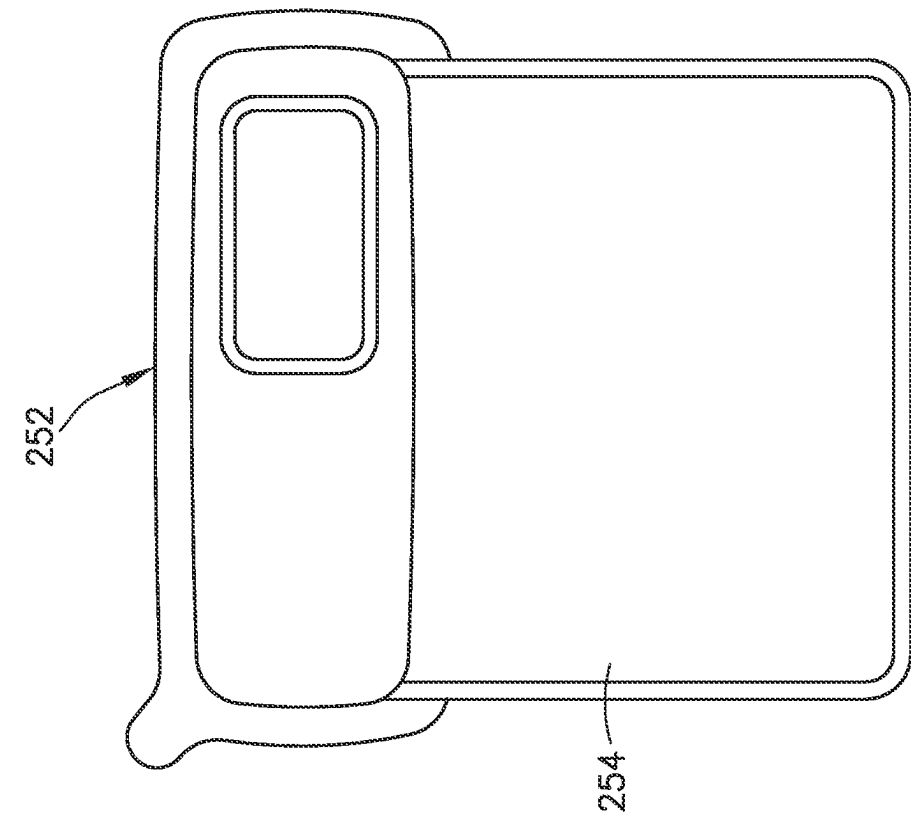


FIG. 37

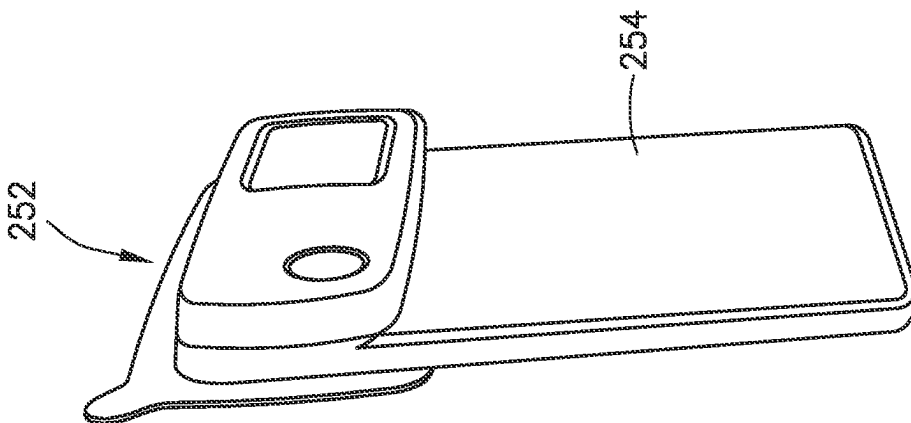


FIG. 36

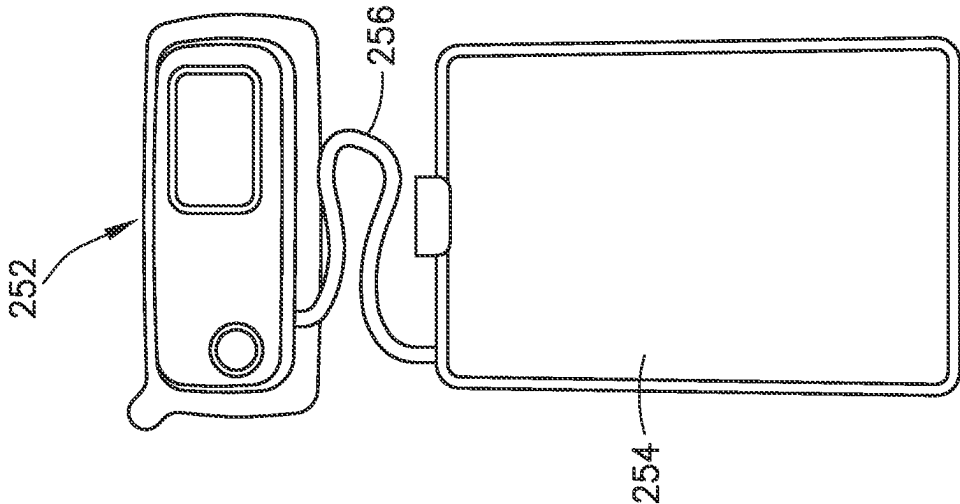


FIG. 38

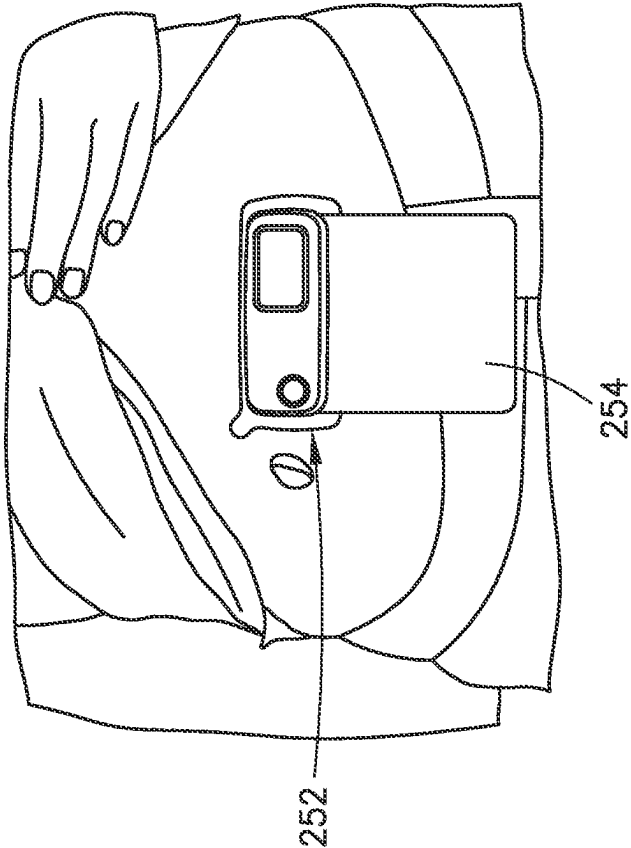


FIG. 39

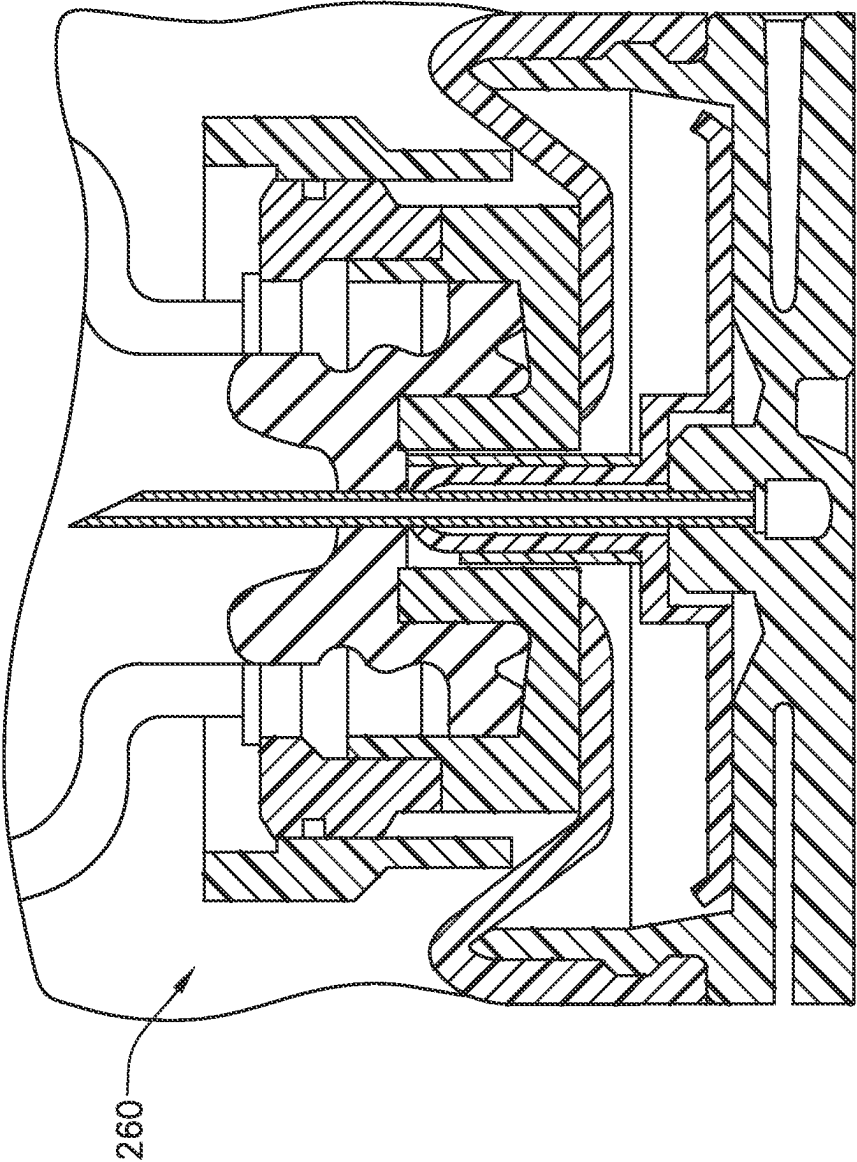


FIG. 40

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NEEDLE HUB FOR DRUG DELIVERY DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority to U.S. Provisional Application No. 63/134,054, filed Jan. 5, 2021, which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

The present disclosure relates to a needle hub for a drug delivery device.

Description of Related Art

Wearable medical devices, such as automatic injectors, have the benefit of providing therapy to the patient at a location remote from a clinical facility and/or while being worn discretely under the patient's clothing. The wearable medical device can be applied to the patient's skin and configured to automatically deliver a dose of a pharmaceutical composition within a predetermined time period after applying the wearable medical device to the patient's skin. After the device delivers the pharmaceutical composition to the patient, the patient may subsequently remove and dispose of the device.

SUMMARY OF THE INVENTION

In one aspect or embodiment, a needle hub for a drug delivery device includes a hub body, an activation button, a needle holder and a needle attached to the needle holder, a cannula holder and a cannula attached to the cannula holder, a needle actuation mechanism configured to move the needle holder and the cannula holder from a retracted position to an insertion position and configured to move the needle holder back to the retracted position, and a cannula spring biasing the cannula holder to a retracted position. The needle actuation mechanism includes a cam track, a cam member received within the cam track, and a torsion spring, with the torsion spring biasing the cam member relative to the cam track. Movement of the activation button is configured to cause the needle holder and the cannula holder to move from the retracted position to the insertion position, with the needle holder configured to return to the retracted position while the cannula holder remains in the insertion position.

The hub body may include a cannula lock configured to lock the cannula holder in the insertion position. The needle hub may include an adhesive pad configured to secure the hub body to a skin surface of a person, with the adhesive pad including a removal tab. Movement of the removal tab is configured to disengage the cannula lock and the hub body to allow the cannula holder to return to the retracted position. The cannula lock may be biased away from the cannula holder via a lock spring, where the hub body includes a hinged portion configured to rotate upon movement of the removal tab and disengage from the cannula lock.

The cannula holder may include a port configured to be in fluid communication with a fluid source, with the cannula in fluid communication with the port. The needle hub may further include tubing connected to the port of the cannula

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holder. The cannula holder may include a seal engaged with the needle. At least a portion of the needle may be received within the cannula.

The needle hub may further include a skin tenting reduction mechanism including an adhesive surface configured to be adhered to a skin surface of a person, with the skin tenting reduction mechanism configured to stretch the skin surface at a location where the needle penetrates the skin surface.

BRIEF DESCRIPTION OF THE DRAWINGS

The above-mentioned and other features and advantages of this disclosure, and the manner of attaining them, will become more apparent and the disclosure itself will be better understood by reference to the following descriptions of embodiments of the disclosure taken in conjunction with the accompanying drawings.

FIG. 1 is a schematic view of a drug delivery device according to one aspect or embodiment of the present application.

FIG. 2 is a schematic view of the drug delivery device of FIG. 1.

FIG. 3 is a perspective view of a needle hub according to one aspect or embodiment of the present application.

FIG. 4A is a top view of the needle hub of FIG. 3, showing an indicator prior to use of the needle hub.

FIG. 4B is a top view of the needle hub of FIG. 3, showing an indicator after insertion of a needle.

FIG. 4C is a top view of the needle hub of FIG. 3, showing an indicator after withdrawal of a cannula.

FIG. 5 is a schematic view showing a method of using the needle hub of FIG. 3.

FIG. 6A is a schematic view of the needle hub of FIG. 3, showing actuation of an activation button.

FIG. 6B is a schematic view of the needle hub of FIG. 3, showing movement of a needle and cannula from a retracted position to an insertion position.

FIG. 6C is a schematic view of the needle hub of FIG. 3, showing a needle in a retracted position and a cannula in an insertion position.

FIG. 6D is a schematic view of the needle hub of FIG. 3, showing movement of a cannula to a retracted position.

FIG. 7 is a perspective view of a needle hub according to a further aspect or embodiment of the present application.

FIG. 8 is a front view of the needle hub of FIG. 7, showing an infusion mode.

FIG. 9 is a front view of the needle hub of FIG. 7, showing a cannula withdrawal.

FIG. 10 is a perspective view of a needle hub according to a further aspect or embodiment of the present application.

FIG. 11 is a schematic view showing a method of using the needle hub of FIG. 7.

FIG. 12A is a schematic view of the needle hub of FIG. 7, showing a pre-use position of the needle hub.

FIG. 12B is a schematic view of the needle hub of FIG. 7, showing the needle hub being actuated.

FIG. 12C is a schematic view of the needle hub of FIG. 7, showing a needle being retracted.

FIG. 12D is a schematic view of the needle hub of FIG. 7, showing a needle in a retracted position.

FIG. 12E is a schematic view of the needle hub of FIG. 7, showing an applicator being detached from the needle hub.

FIG. 12F is a schematic view of the needle hub of FIG. 7, showing a cannula in a retracted position.

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FIG. 13 is a cross-sectional view of a needle hub according to a further aspect or embodiment of the present application.

FIG. 14 is a cross-sectional view of the needle hub of FIG. 13.

FIG. 15 is a perspective view of a needle hub according to a further aspect or embodiment of the present application.

FIG. 16 is a perspective view of a needle hub according to a further aspect or embodiment of the present application.

FIG. 17 is a perspective view of a needle actuation assembly according to one aspect or embodiment of the present application.

FIG. 18A is a cross-sectional view of the needle hub of FIG. 15, showing a cannula insertion position.

FIG. 18B is a cross-sectional view of the needle hub of FIG. 15, showing a cannula retraction position.

FIG. 19 is a perspective view of a needle hub according to a further aspect or embodiment of the present application.

FIG. 20 is a schematic view showing a method of using the needle hub of FIG. 19.

FIG. 21 is a perspective view of a needle actuation assembly according to a further aspect or embodiment of the present application.

FIG. 22 is a perspective view of a drug delivery device and a needle hub according to a further aspect or embodiment of the present application.

FIG. 23 is an exploded perspective view of the drug delivery device and the needle hub of FIG. 22.

FIG. 24 is a perspective view of the drug delivery device and the needle hub of FIG. 22, showing the drug delivery device and the needle hub connected while delivering a medicament.

FIG. 25 is a perspective view of the drug delivery device and the needle hub of FIG. 22, showing the needle hub separated from the drug delivery device while delivering a medicament.

FIG. 26A is a cross-sectional view of a needle hub according to a further aspect or embodiment of the present application, showing an initial position of the needle hub.

FIG. 26B is a cross-sectional view of a needle hub according to a further aspect or embodiment of the present application, showing engagement with a skin surface.

FIG. 26C is a cross-sectional view of a needle hub according to a further aspect or embodiment of the present application, showing insertion of a needle.

FIG. 27A is a cross-sectional view of a needle hub according to a further aspect or embodiment of the present application, showing an initial position of the needle hub.

FIG. 27B is a cross-sectional view of a needle hub according to a further aspect or embodiment of the present application, showing engagement with a skin surface.

FIG. 27C is a cross-sectional view of a needle hub according to a further aspect or embodiment of the present application, showing insertion of a needle.

FIG. 28A is a cross-sectional view of a needle hub according to a further aspect or embodiment of the present application, showing an initial position of the needle hub.

FIG. 28B is a cross-sectional view of a needle hub according to a further aspect or embodiment of the present application, showing engagement with a skin surface.

FIG. 28C is a cross-sectional view of a needle hub according to a further aspect or embodiment of the present application, showing insertion of a needle.

FIG. 29 is a perspective view of a needle actuation assembly according to a further aspect or embodiment of the present application, showing an unactuated position.

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FIG. 30 is a perspective view of the needle actuation assembly of FIG. 29, showing an actuated position.

FIG. 31 is a cross-sectional view of the needle actuation assembly of FIG. 29, showing an unactuated position.

FIG. 32 is a cross-sectional view of the needle actuation assembly of FIG. 29, showing an actuated position.

FIG. 33 is a cross-sectional view of the needle actuation assembly of FIG. 29, showing a retracted position.

FIG. 34 is a perspective view of a needle hub according to a further aspect or embodiment of the present application.

FIG. 35 is a schematic view of the needle hub of FIG. 34.

FIG. 36 is a perspective view of a drug delivery device according to a further aspect or embodiment of the present application.

FIG. 37 is a front view of the drug delivery device of FIG. 36.

FIG. 38 is a front view of the drug delivery device of FIG. 36, showing a reservoir separated from the drug delivery device.

FIG. 39 is a front view of the drug delivery device of FIG. 36, showing the drug delivery device attached to a patient.

FIG. 40 is a partial cross-sectional view of a prior art valve assembly.

Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate exemplary embodiments of the disclosure, and such exemplifications are not to be construed as limiting the scope of the disclosure in any manner.

DETAILED DESCRIPTION OF THE INVENTION

Spatial or directional terms, such as “left”, “right”, “inner”, “outer”, “above”, “below”, and the like, are not to be considered as limiting as the invention can assume various alternative orientations.

All numbers used in the specification and claims are to be understood as being modified in all instances by the term “about”. By “about” is meant a range of plus or minus ten percent of the stated value. As used in the specification and the claims, the singular form of “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise. The terms “first”, “second”, and the like are not intended to refer to any particular order or chronology, but instead refer to different conditions, properties, or elements. By “at least” is meant “greater than or equal to”.

Referring to FIGS. 1-3, a drug delivery device 10 includes a reservoir 12, a power module 14, an insertion mechanism 16, control electronics 18, and a housing 20. In one aspect or embodiment, the drug delivery device 10 is a wearable automatic injector. The drug delivery device 10 may be mounted onto the skin of a patient and triggered to inject a pharmaceutical composition from the reservoir 12 into the patient. The drug delivery device 10 may be pre-filled with the pharmaceutical composition, or it may be filled with the pharmaceutical composition by the patient or medical professional prior to use. The control electronics 18 may include a processor 22, such as a microcontroller, a motor driver 23, a sensing module 24, a visual driver 25, and/or audio driver 26. The drug delivery device 10 includes a drive mechanism 27 configured to dispense fluid from the reservoir 12. The drive mechanism 27 may be motor powered, spring powered, hydraulic powered, pneumatic powered, and/or other suitable drive mechanism.

The drug delivery device 10 is configured to deliver a dose of a pharmaceutical composition, e.g., any desired medicament, into the patient's body by a subcutaneous

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injection at a slow, controlled injection rate. Exemplary time durations for the delivery achieved by the drug delivery device 10 may range from about 5 minutes to about 60 minutes, but are not limited to this exemplary range. Exemplary volumes of the pharmaceutical composition delivered by the drug delivery device 10 may range from about 10 milliliters to about 50 milliliters, but are not limited to this exemplary range. The volume of the pharmaceutical composition delivered to the patient may be adjusted. The device 10 may communicate with another device, such as a mobile device or computer.

Referring to FIGS. 3-6D, according to one aspect or embodiment, the insertion mechanism 16 includes a needle hub 30 separate from the housing 20. The needle hub 30 includes a removal tab 32, an activation button 34, a status indicator 36, a finger grip, side grips, and an integrated cannula withdrawal tab 38. The needle hub 30 includes an in-dwelling cannula. The status indicator 36 may be white when unused (FIG. 4A), blue when the needle has been inserted and ready to infuse (FIG. 4B), and green when the cannula has been withdrawn (FIG. 4C). As shown in FIG. 5, the needle hub 30 is used by removing the packaging, removing an adhesive liner from the bottom of the needle hub 30, attaching the needle hub 30 to a skin surface, and squeezing the activation button 34, which causes the needle to automatically retract leaving an in-dwelling cannula in the patient. The medicament or fluid is then infused into the patient. Once the infusion is complete, the cannula withdrawal tab 38 is pulled, which retracts the cannula. The needle hub 30 can then be removed from the skin of the patient.

Referring to FIGS. 6A-6D, in one aspect or embodiment, the needle hub 30 includes a hub body 42, the activation button 34, a needle holder 44, a needle 46 attached to the needle holder 44, a needle spring 48, a cannula holder 50, a cannula 52 attached to the cannula holder 50, and a cannula spring 54. The activation button 34 is moveable relative to the hub body 42 and has a first actuation surface 56. The needle holder 44 is moveable relative to the hub body 42 and has a second actuation surface 58. The first actuation surface 56 of the activation button 34 is configured to engage the second actuation surface 58 of the needle holder 44. The needle spring 48 biases the needle holder 44 to a retracted position where the needle 46 is positioned within the hub body 42. The cannula holder 50 is moveable relative to the hub body 42 and the needle holder 44. The cannula 52 is configured to be in fluid communication with a fluid source, such as the fluid reservoir 12. The cannula spring 54 biases the cannula holder 50 to a retracted position where the cannula 52 is positioned within the hub body 42. Movement of the activation button 34 is configured to cause the first actuation surface 56 of the activation button 34 to engage the second actuation surface 58 of the needle holder 44 to move the needle holder 44 and the cannula holder 50 from the respective retracted positions to insertion positions where distal ends of the needle 46 and the cannula 52 are positioned outside of the hub body 42, with the needle holder 44 configured to return to the retracted position while the cannula holder 50 remains in the insertion position. Movement of a portion of the hub body 42 is configured to disengage a connection between the cannula holder 50 and the hub body 42 to allow the cannula holder 50 to return to the retracted position.

Referring again to FIGS. 6A-6D, the needle holder 44 includes a passageway 60 configured to be in fluid communication with the fluid reservoir 12, with the needle 46 in fluid communication with the passageway 60 of the needle

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holder 44. At least a portion of the needle 46 is received within the cannula 52. Fluid is configured to flow from the fluid reservoir 12 via tubing 62 to the passageway 60 of the needle holder 44, through the needle 46, and into the cannula 52. The cannula holder 50 includes a seal 64 engaged with the needle 46. The needle hub 30 includes a first projection 66 and the cannula holder 50 includes a second projection 68, with the first projection 66 of the needle hub 66 engaging the second projection 68 of the cannula holder 50 when the cannula 52 is in the insertion position to restrict movement of the cannula holder 50 to the retracted position.

Referring to FIGS. 3-6D, the needle hub 30 includes a removal tab 70, where movement of the removal tab 70 releases an engagement between the first projection 66 of the needle hub 30 and the second projection 68 of the cannula holder 50 to allow the cannula spring 54 to bias the cannula 52 to the retracted position. At least a portion of the removal tab 70 is configured to be engaged with a skin surface of a person after attaching the needle hub 30 to a person. The activation button 34 is moveable along a first axis 72, and the needle holder 44 and the cannula holder 50 are moveable along a second axis 74 perpendicular to the first axis 72. The first actuation surface 56 of the activation button 34 is configured to disengage from the second actuation surface 58 of the needle holder 44 after movement of the activation button 34 is a predetermined distance along the first axis 72.

Referring to FIGS. 7-14, a needle hub 80 according to a further aspect or embodiment includes an applicator 82 having a needle holder 84, a needle 86 attached to the needle holder 84, a needle retraction spring 88, and an activation button 90, and a hub body 92 having a cannula holder 94, a cannula 96 attached to the cannula holder 94, a cannula withdrawal button 98, and a cannula retraction spring 100. At least a portion of the hub body 92 is configured to be received within the applicator 82 and the applicator 82 is configured to be separated from the hub body 92. Movement of the activation button 90 is configured to move the needle holder 84 and the cannula holder 94 from a retracted position, where the needle 86 and the cannula 96 are positioned within the applicator 82 or hub body 92, to an insertion position, where distal ends of the needle 86 and the cannula 96 are positioned outside of the applicator 82 and the hub body 92. The cannula withdrawal button 98 locks the cannula holder 94 in the insertion position against a biasing force of the cannula retraction spring 100 when the cannula holder 94 is moved from the retracted position to the insertion position. As shown in FIG. 10, in one aspect or embodiment, the cannula withdrawal button 98 may be omitted.

Referring to FIGS. 12A-12F, the needle holder 84 is configured to move to the retracted position after movement of the cannula holder 94 to the insertion position. The activation button 90 includes an extension 102 having a drive protrusion 103 and the applicator 82 includes a drive surface 104 configured to engage the drive protrusion 103. Upon movement of the activation button 90, the drive protrusion 103 engages the needle holder 84 to move the needle holder 84 and the cannula holder 94 to the insertion position, with the drive protrusion 103 engaging the drive surface 104 of the applicator 82 to move the extension 102 radially outward thereby releasing the needle holder 84 from the drive protrusion 103 to allow the needle retraction spring 100 to return the needle holder 84 to the retracted position. Actuation of the cannula withdrawal button 98 is configured to move the cannula holder 94 from the insertion position to the retracted position. The hub body 92 further includes an adhesive pad 105 configured to secure the hub body 92 to a

skin surface of a person. The adhesive pad **105** includes a removal tab **106** extending radially outward from the hub body **92**. The activation button **90** is received within an opening **107** defined by a body **108** of the applicator **82**. The cannula holder **94** includes a port **109** configured to be in fluid communication with the fluid reservoir **12**, with the cannula **96** in fluid communication with the port **109**. Tubing **110** is connected to the port of the cannula holder **94**.

Referring to FIG. **11**, in one aspect or embodiment, the needle hub **80** is used by removing the packaging, removing an adhesive liner, attaching the needle hub **80** to a skin surface of a patient and removing a safety cap, and pressing the activation button **90** of the applicator **82**, which automatically actuates and retracts the needle **86** to leave the in-dwelling cannula **96**. The applicator **82** can then be removed from the hub body **92** and the infusion can commence. Once the infusion is complete, the cannula withdrawal button **98** may be pressed to remove the cannula **96** from the patient, with the hub body **92** being removed from the skin of the patient using the removal tab **106**.

Referring to FIGS. **13** and **14**, in one aspect or embodiment, the cannula holder **94** includes a portion of the adhesive pad **105**, which removes a portion of the adhesive pad **105** from the skin of the patient when the cannula holder **94** is removed from the hub body **92** to facilitate easier removal of the remainder of the adhesive pad **105** from the skin of the patient.

Referring to FIGS. **15-18B**, a needle hub **112**, according to a further aspect or embodiment, includes a hub body **114**, an activation button **116**, a needle holder **118** and a needle **120** attached to the needle holder **118**, a cannula holder **122** and a cannula **124** attached to the cannula holder **122**, a needle actuation mechanism **126**, and a cannula spring **128**. The needle actuation mechanism **126** is configured to move the needle holder **118** and the cannula holder **122** from a retracted position to an insertion position and is configured to move the needle holder **118** back to the retracted position. The needle actuation mechanism **126** includes a cam track **130**, a cam member **132** received within the cam track **130**, and a torsion spring **134**. The torsion spring **134** biases the cam member **132** relative to the cam track **130**. The cannula spring **128** biases the cannula holder **122** to a retracted position. Movement of the activation button **116** is configured to cause the needle holder **118** and the cannula holder **122** to move from the retracted position to the insertion position, with the needle holder **118** configured to return to the retracted position while the cannula holder **122** remains in the insertion position. As shown in FIG. **16**, in one aspect or embodiment, the needle hub **112** includes two lateral activation squeeze buttons **116**. The needle hub **112** is used in the same manner as described above in connection with the needle hub **30** shown in FIG. **5**.

Referring to FIGS. **17-18B**, the hub body **114** includes a cannula lock **136** configured to lock the cannula holder **122** in the insertion position. The needle hub **112** includes an adhesive pad **138** configured to secure the hub body **114** to a skin surface of a person, with the adhesive pad **138** including a removal tab **140**. Movement of the removal tab **140** is configured to disengage the cannula lock **136** and the hub body **114** to allow the cannula holder **122** to return to the retracted position. The cannula lock **136** is biased away from the cannula holder **122** via a lock spring **142**, where the hub body **112** includes a hinged portion **144**, with the hinged portion **144** configured to rotate upon movement of the removal tab **140** and disengage from the cannula lock **136**. The cannula holder **122** includes a port **146** configured to be in fluid communication with the fluid reservoir **12**, with the

cannula **124** in fluid communication with the port **146**. The needle hub **112** includes tubing **148** connected to the port **146** of the cannula holder **122**. The cannula holder **122** includes a seal **150** engaged with the needle **120**, with at least a portion of the needle **120** received within the cannula **124**.

Referring to FIGS. **19-21**, a needle hub **152** according to a further aspect or embodiment includes a needle holder **154** and a needle **156** attached to the needle holder **154**, a needle actuation assembly **158** configured to move the needle holder **154** from a retracted position, to an insertion position, and back to the retracted position, and a pressure interlock **160** including an inlet **162** configured to be in fluid communication with the fluid reservoir **12**, an outlet **164** in fluid communication with the needle **156**, and a lock member **166**. The lock member **166** has a first position where the lock member **166** prevents actuation of the needle actuation assembly **158** and a second position where the lock member **166** allows actuation of the needle actuation assembly **158**. The lock member **166** is moved from the first position to the second position based on a pressure within the pressure interlock **160**.

Referring to FIG. **21**, the lock member **166** isolates the inlet **162** from the outlet **164** when the lock member **166** is in the first position, and the lock member **166** allows fluid communication between the inlet **162** and the outlet **164** when the lock member **166** is in the second position. The needle hub **152** further includes tubing **168** connected to the outlet **164** and in fluid communication with the needle **156**. The lock member **166** comprises an opening **170**, with a portion of the needle actuation assembly **158** extending through the opening **170** of the lock member **166** when the lock member **166** is in the second position. The needle actuation assembly **158** includes a cam track **172**, a cam member **174**, and an actuation spring **176** biasing the cam member **174** relative to the cam track **172**. A cam block **178** defines the cam track **172**, with the cam block **178** extending through the opening **170** of the lock member **166** when the lock member **166** is in the second position. The needle actuation assembly **158** further includes a cannula **180**, where the needle **156** is received within the cannula **180** when the needle holder **156** is in the retracted position.

Referring again to FIGS. **19-21**, the needle hub **152** includes a housing **182** and a removal tab **184**. A top surface **186** of the housing **182** is smooth and free of activation buttons. As shown in FIG. **20**, the needle hub **152** is used by removing packaging, removing an adhesive liner, attaching the needle hub **152** to a skin surface of a patient, and activating the drive mechanism **27** to insert the needle **156**, with the needle **156** automatically retracting leaving the in-dwelling cannula **180**. After infusion is complete, the needle hub **152** is removed by grasping the removal tab **184** and lifting upwards, with the cannula **180** automatically retracting. In one aspect or embodiment, pulling the removal tab **184** causes a drop in pressure of the pressure interlock **160** to cause the cannula holder and/or the cannula **180** to automatically retract.

Referring to FIGS. **22-25**, a drug delivery device **190** and a needle hub **192**, according to a further aspect or embodiment, is shown. The drug delivery device **190** may be similar to the drug delivery device **10** shown in FIGS. **1** and **2**. The drug delivery device **190** and the needle hub **192** of FIGS. **22-25**, however, is modular, with the needle hub **192** optionally integrated within the drug delivery device **190** (FIG. **24**) or with the needle hub **192** separated from the drug delivery device **190** and separately attached to a skin surface of a patient (FIG. **25**). In one aspect or embodiment, the drug

delivery device **190** and the needle hub **192** may remain connected or integral for lower drug volume and separated with a fluid connection therebetween for larger drug volumes.

Referring to FIGS. **26A-26C**, a needle hub **200** with a skin tenting reduction feature according to one aspect or embodiment is shown. The needle hub **200** includes a rotating engagement mechanism **202**, with a portion of the rotating engagement mechanism **202** first contacting a skin surface of the patient and adhering to the skin surface and further rotating as the needle hub **200** is fully pressed onto the skin surface. The initial adherence and further rotation of the rotating engagement mechanism stretches the skin to reduce skin tenting.

Referring to FIGS. **27A-27C**, a needle hub **204** with a skin tenting reduction feature according to one aspect or embodiment is shown. The needle hub **204** includes an adhesive ring **206** that is pressed onto the skin prior to insertion of a needle when an activation button is depressed or actuated. The adhesive ring **206** stretches the skin to reduce skin tenting.

Referring to FIGS. **28A-28C**, a needle hub **208** with a skin tenting reduction feature according to one aspect or embodiment is shown. The needle hub **208** includes skin stretching member **210** that is moved radially outward after being initially adhered to a skin surface of a patient. An activation button **212** engages the skin stretching member **210** to move the skin stretching member **210** radially outward, which stretches the skin locally to reduce skin tenting.

The skin tenting reduction features and associated mechanisms of FIGS. **26A-28C** may be incorporated into any of the aspect or embodiments of the needle hub or needle insertion arrangements disclosed herein.

Referring to FIGS. **29-33**, a needle actuation assembly **220**, according to one aspect or embodiment, includes a clip **222** that holds a needle actuator body **224** and cannula body **226** in the retracted position, which are biased by a spring **228**. Pushing the clip **222** inwards releases the needle actuator body **224** and cannula body **226** to cause insertion of a needle **230** and a cannula **232**. When the cannula body **226** reaches the bottom of a housing **233**, the cannula body **226** contacts angled features causing the cannula body **226** to rotate and/or twist. The cannula body **226** is held down by clips **236** in the walls of the housing **233**. After the cannula body **226** rotates and/or twists, the needle actuator body **224** is released and a return spring **238** retracts the needle **230**.

Referring to FIGS. **34** and **35**, a needle hub **240** according to a further aspect or embodiment is configured to be decoupled from remaining components of a drug delivery device. The needle hub **240** includes a protective cap **242**, a needle insertion mechanism **244**, connection arrangement **246** configured to place the needle hub **240** in fluid communication with the reservoir **12** and the drive mechanism **27**, a fluid path **248**, and an adhesive pad and/or layer **250**. The connection arrangement **246** may provide for aseptic connection between the needle hub **240** and the reservoir **12**. The needle retraction may be manually activated or automatically activated via a triggering mechanism connected to an end of dose event and/or a wireless connection between the driving unit and the needle hub **240**. The triggering mechanism may include a flexible rigid connection to plunger rod movement (totally or partially at the end of translation). The fluid path **248** and connection arrangement **246** is maintained sterile until the connection is established, with sterilization of the sub-system and reservoir.

Referring to FIGS. **36-39**, a drug delivery device **252**, according to a further aspect or embodiment, includes a

flexible reservoir **254**, with at least a portion of the flexible reservoir **254** positioned externally from a remaining portion of the drug delivery device **252**. The drug delivery device **252** may be similar to the drug delivery device **10** shown in FIGS. **1** and **2**. As shown in FIG. **39**, for smaller volumes, such as 10 mL-30 mL, the flexible reservoir **254** may be directly attached to the drug delivery device **252** and worn on a skin surface of the patient. As shown in FIG. **38**, for larger volumes, such as 50 mL, the flexible reservoir **254** may be separated from the drug delivery device **252** and fluidly connected to the drug delivery device **252** via a fluid path **256**, such as a tube. The flexible reservoir **254** may be separately attached to the patient via a belt clip, harness, strap, or other suitable arrangement.

Referring to FIG. **40**, the drug delivery devices in any of the aspects or embodiments discussed above may utilize a valve assembly **260** that engages a reservoir and/or container to facilitate the fluid connection between the reservoir and/or container and the fluid path to the needle and/or cannula. The valve assembly **260** may be similar to and operate in the same manner as the valve assembly shown and described in U.S. Patent Application Publication No. 2017/0354788.

Although the invention has been described in detail for the purpose of illustration based on what is currently considered to be the most practical and preferred embodiments, it is to be understood that such detail is solely for that purpose and that the invention is not limited to the disclosed embodiments, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. For example, it is to be understood that the present invention contemplates that, to the extent possible, one or more features of any embodiment can be combined with one or more features of any other embodiment.

The invention claimed is:

1. A needle hub for a drug delivery device comprising:

- a hub body;
- an activation button;
- a needle holder and a needle attached to the needle holder;
- a cannula holder and a cannula attached to the cannula holder;
- a needle actuation mechanism configured to move the needle holder and the cannula holder from a retracted position to an insertion position and configured to move the needle holder back to the retracted position, the needle actuation mechanism comprising a cam track, a cam member received within the cam track, and a torsion spring, the torsion spring biasing the cam member relative to the cam track; and
- a cannula spring biasing the cannula holder to the retracted position,

wherein movement of the activation button is configured to cause the needle holder and the cannula holder to move from the retracted position to the insertion position, with the needle holder configured to return to the retracted position while the cannula holder remains in the insertion position,

wherein the hub body comprises a cannula lock configured to lock the cannula holder in the insertion position, the needle hub further comprising an adhesive pad configured to secure the hub body to a skin surface of a person, and the adhesive pad comprising a removal tab, wherein movement of the removal tab is configured to disengage the cannula lock and the hub body to allow the cannula holder to return to the retracted position, and

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wherein the cannula lock is biased away from the cannula holder via a lock spring, and wherein the hub body comprises a hinged portion, the hinged portion configured to rotate upon movement of the removal tab and disengage from the cannula lock.

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2. The needle hub of claim 1, wherein the cannula holder comprises a port configured to be in fluid communication with a fluid source, the cannula being in fluid communication with the port.

3. The needle hub of claim 2, further comprising tubing 10 connected to the port of the cannula holder.

4. The needle hub of claim 3, wherein the cannula holder comprises a seal engaged with the needle.

5. The needle hub of claim 4, wherein at least a portion of the needle is received within the cannula.

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